MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL,

BINARY FULL ADDERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, full adder microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit
01	2-Bit full adder
02	4-Bit full adder
03	Dual full adder
04	Gated full adder

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	MIL-M-38510, appendix C case outline
A	F-1 (14-pin, 1/4" x 1/4", flat-pack)
B	F-3 (14-pin, 1/4" x 1/8", flat-pack)
C	D-1 (14-pin, 1/4" x 3/4", dual-in-line)
D	F-2 (14-pin, 1/4" x 3/8", flat-pack)
E	D-2 (16-pin, 1/4" x 7/8", dual-in-line)
F	F-5 (16-pin, 1/4" x 3/8", flat-pack)

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 Vdc to 7.0 Vdc
Input voltage range	-1.5 Vdc at -12 mA to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Maximum power dissipation 1/	P_D = 275 mWdc for types 01 and 03 P_D = 550 mWdc for type 02 P_D = 165 mWdc for type 04
Lead temp. (soldering 10 secs)	300°C
Thermal resistance, junction to case	θ_{JC} = 0.09°C/mW for flat pack 0.08°C/mW for dual-in-line pack
Junction temperature	T_J = 175°C

1/ Must withstand the added P_D due to short circuit condition (e.g. I_{OS}) at one output for five (5) seconds duration.

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1.4 Recommended operating conditions.

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc
Normalized fanout (each input) 1/	10 maximum for sum outputs 5 maximum for carry outputs
Ambient operating temperature range	-55°C to 125°C

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510 and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. Case outlines shall be as specified in 1.2.3.

3.2.2 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.3 Truth tables. The truth tables shall be as specified on figure 2.

3.2.4 Logic diagrams. The logic diagram shall be as specified on figure 3.

3.2.5 Schematic circuit. The schematic circuit shall be as specified on figure 4.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and apply over the full recommended ambient operating temperature range, unless otherwise specified.

1/ Device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 0.8 V or 2.0 V 3/	All	2.4	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IN} = 2.0 V or 0.8 V 4/	All	---	0.4	Volts
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA, T_A = 25°C	All	---	-1.5	Volts
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 2/				
		A_1, B_1, A_2, B_2, C_0	01	---	160	µA
		A_1, B_1, A_3, B_3, C_0	02	---	160	µA
		A_2, B_2, A_4, B_4	02	---	40	µA
		A_2, B_2	03	---	40	µA
		$A_1, A_2, B_1, B_2, A_C, B_C$	04	---	15	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 2/	01, 02	---	400	µA
High-level input current	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 2/				
		$A_1, B_1, \overline{A}_2, B_2, C_1, \overline{C}_2$	03	---	160	µA
		C_N	04	---	200	µA
High-level input current	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 2/	03	---	400	µA
			04	---	1	mA
Low-level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 1/				
		A_1, B_1, C_0	01	-1.4	-6.4	mA
		A_1, B_1, A_3, B_3, C_0	02	-0.8	-6.4	mA
		A_2, B_2	03	-0.4	-1.3	mA
		$A_1, A_2, B_1, B_2, A_C, B_C$	04	---	-1.6	mA
Low-level input current	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 1/				
		A_2, B_2	01	-0.7	-3.2	mA
		A_2, B_2, A_4, B_4	02	-0.7	-3.2	mA
		A_1, B_1, C_1	03	-1.2	-5.6	mA
		A^*, B^*	04	---	-2.8	mA
Low-level input current	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V 1/				
		A_2, B_2, C_2	03	-1.2	-5.6	mA
		C_N	04	---	-8.0	mA
Short-circuit output current at Σ	I_{OS1}	V_{CC} = 5.5 V, V_{IN} = 4.5 V, V_{OUT} = 0 V 5/	01	-20	-70	mA
			02	-20	-55	mA
			03	-30	-100	mA
			04	-20	-57	mA
Short-circuit output current at C	I_{OS2}	V_{CC} = 5.5 V, V_{IN} = 4.5 V, V_{OUT} = 0 V 5/	01, 02, 04	-20	-70	mA
Short-circuit output current at A* or B*	I_{OS3}	V_{CC} = 5.5 V, V_{IN} = 0 V, V_{OUT} = 0 V 5/	04	-0.9	-2.9	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 4.5 V	01	---	50	mA
			02	---	100	mA
			03	---	55	mA
			04	---	31	mA
Propagation delay time, high-to-low-level, C_0 to Σ_1	t_{PHL1}	C_L = 50 pF, R_L = 390 Ω	01	3	61	ns
			02	3	51	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics. - Continued

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
Propagation delay time, low-to-high-level, C_0 to Σ_1	t_{PLH1}	C_L = 50 pF, R_L = 390 Ω	01	3	56	ns
			02	3	39	
Propagation delay time, high-to-low-level, B_2 to Σ_2	t_{PHL2}	C_L = 50 pF, R_L = 390 Ω	01	3	54	ns
			02	3	40	
Propagation delay time, low-to-high-level, B_2 to Σ_2	t_{PLH2}	C_L = 50 pF, R_L = 390 Ω	01	3	65	ns
			02	3	45	
Propagation delay time, high-to-low-level, C_0 to Σ_2	t_{PHL3}	C_L = 50 pF, R_L = 390 Ω	01	3	61	ns
			02	3	54	
Propagation delay time, low-to-high-level, C_0 to Σ_2	t_{PLH3}	C_L = 50 pF, R_L = 390 Ω	01	3	62	ns
			02	3	51	
Propagation delay time, high-to-low-level, C_0 to C_2	t_{PHL4}	C_L = 50 pF, R_L = 800 Ω	01	3	43	ns
Propagation delay time, low-to-high-level, C_0 to C_2	t_{PLH4}	C_L = 50 pF, R_L = 800 Ω	01	3	34	ns
Propagation delay time, high-to-low-level, C_0 to Σ_3	t_{PHL4}	C_L = 50 pF, R_L = 390 Ω	02	3	77	ns
Propagation delay time, low-to-high-level, C_0 to Σ_3	t_{PLH4}	C_L = 50 pF, R_L = 390 Ω	02	3	62	ns
Propagation delay time, high-to-low-level, C_0 to Σ_4	t_{PHL5}	C_L = 50 pF, R_L = 390 Ω	02	3	71	ns
Propagation delay time, low-to-high-level, C_0 to Σ_4	t_{PLH5}	C_L = 50 pF, R_L = 390 Ω	02	3	72	ns
Propagation delay time, high-to-low-level, C_0 to C_4	t_{PHL6}	C_L = 50 pF, R_L = 800 Ω	02	3	55	ns
Propagation delay time, low-to-high-level, C_0 to C_4	t_{PLH6}	C_L = 50 pF, R_L = 800 Ω	02	3	63	ns
Propagation delay time, high-to-low-level, A_2 to Σ_2	t_{PHL7}	C_L = 50 pF, R_L = 390 Ω	02	3	40	ns
Propagation delay time, low-to-high-level, A_2 to Σ_2	t_{PLH7}	C_L = 50 pF, R_L = 390 Ω	02	3	45	ns
Propagation delay time, high-to-low-level, A_4 to Σ_4	t_{PHL8}	C_L = 50 pF, R_L = 390 Ω	02	3	40	ns
Propagation delay time, low-to-high-level, A_4 to Σ_4	t_{PLH8}	C_L = 50 pF, R_L = 390 Ω	02	3	45	ns
Propagation delay time, high-to-low-level, B_4 to Σ_4	t_{PHL9}	C_L = 50 pF, R_L = 390 Ω	02	3	40	ns
Propagation delay time, low-to-high-level, B_4 to Σ_4	t_{PLH9}	C_L = 50 pF, R_L = 390 Ω	02	3	45	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics. - Continued

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Unit
Propagation delay time, high-to-low-level, C_1 to Σ_1	t_{PHL1}	C_L = 50 pF, R_L = 390 Ω	03	9	59	ns
Propagation delay time, low-to-high-level, C_1 to Σ_1	t_{PLH1}	C_L = 50 pF, R_L = 390 Ω	03	9	60	ns
Propagation delay time, high-to-low-level, C_1 to $\overline{\Sigma}_1$	t_{PHL2}	C_L = 50 pF, R_L = 390 Ω	03	6	39	ns
Propagation delay time, low-to-high-level, C_1 to $\overline{\Sigma}_1$	t_{PLH2}	C_L = 50 pF, R_L = 390 Ω	03	6	39	ns
Propagation delay time, high-to-low-level, C_1 to $\overline{C}_{01}$	t_{PHL3}	C_L = 50 pF, R_L = 800 Ω	03	3	21	ns
Propagation delay time, low-to-high-level, C_1 to $\overline{C}_{01}$	t_{PLH3}	C_L = 50 pF, R_L = 800 Ω	03	3	21	ns
Propagation delay time, high-to-low-level, A_2 to $\overline{\Sigma}_2$	t_{PHL4}	C_L = 50 pF, R_L = 390 Ω	03	11	67	ns
Propagation delay time, low-to-high-level, A_2 to $\overline{\Sigma}_2$	t_{PLH4}	C_L = 50 pF, R_L = 390 Ω	03	11	69	ns
Propagation delay time, high-to-low-level, A_2 to Σ_2	t_{PHL5}	C_L = 50 pF, R_L = 390 Ω	03	8	48	ns
Propagation delay time, low-to-high-level, A_2 to Σ_2	t_{PLH5}	C_L = 50 pF, R_L = 390 Ω	03	8	49	ns
Propagation delay time, high-to-low-level, A_2 to $\overline{C}_{02}$	t_{PHL6}	C_L = 50 pF, R_L = 800 Ω	03	5	28	ns
Propagation delay time, low-to-high-level, A_2 to $\overline{C}_{02}$	t_{PLH6}	C_L = 50 pF, R_L = 800 Ω	03	5	29	ns
Propagation delay time, high-to-low-level, C_n to $\overline{C}_{n+1}$	t_{PHL1}	C_L = 50 pF, R_L = 780 Ω	04	---	18	ns
Propagation delay time, low-to-high-level, C_n to $\overline{C}_{n+1}$	t_{PLH1}	C_L = 50 pF, R_L = 780 Ω	04	---	33	ns
Propagation delay time, high-to-low-level, B_C to $\overline{C}_{n+1}$	t_{PHL2}	C_L = 50 pF, R_L = 780 Ω	04	---	90	ns
Propagation delay time, low-to-high-level, B_C to $\overline{C}_{n+1}$	t_{PLH2}	C_L = 50 pF, R_L = 780 Ω	04	---	45	ns
Propagation delay time, high-to-low-level, A_C to Σ	t_{PHL3}	C_L = 50 pF, R_L = 400 Ω	04	---	127	ns
Propagation delay time, low-to-high-level, A_C to Σ	t_{PLH3}	C_L = 50 pF, R_L = 400 Ω	04	---	112	ns
Propagation delay time, high-to-low-level, B_C to $\overline{\Sigma}$	t_{PHL4}	C_L = 50 pF, R_L = 400 Ω	04	---	120	ns
Propagation delay time, low-to-high-level, B_C to $\overline{\Sigma}$	t_{PLH4}	C_L = 50 pF, R_L = 400 Ω	04	---	90	ns
Propagation delay time, high-to-low-level, A_1 to A*	t_{PHL5}	C_L = 50 pF	04	---	45	ns
Propagation delay time, low-to-high-level, A_1 to A*	t_{PLH5}	C_L = 50 pF	04	---	105	ns
Propagation delay time, high-to-low-level, B_1 to B*	t_{PHL6}	C_L = 50 pF	04	---	45	ns
Propagation delay time, low-to-high-level, B_1 to B*	t_{PLH6}	C_L = 50 pF	04	---	105	ns

See footnotes on page 6.

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1/ All unspecified inputs at 5.5 volts.
2/ All unspecified inputs grounded.
3/ I_{OH} for each sum output is -0.4 mA (types 01, 02 and 04) and -0.8 mA (type 03) and for each carry output it is -0.2 mA (types 01, 02 and 04) and -0.56 mA (type 03). I_{OH} for A* or B* is -0.12 mA (type 04).
4/ I_{OL} for each sum output is 16 mA and for each carry output it is 8 mA. I_{OL} for A* or B* is 4.8 mA (type 04).
5/ Not more than one output should be shorted at one time.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification and quality conformance, by device class are specified in table II. (Subgroups 7 and 8 testing requires only a summary of attributes data.)

TABLE II. Electrical test requirements.

MIL-STD-883 Test requirement	Subgroups (see table III)		
	Class A Devices	Class B Devices	Class C Devices
Interim electrical parameters (Pre Burn-In) (Method 5004)	1	1	None
Final electrical test parameters (Method 5004)	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9	1, 7
Group A test requirements (Method 5005)	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 9 10, 11	1, 2, 3, 7, 9
Groups C and D end point electrical parameters (Method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for Group C periodic inspections	None	None	10, 11

* PDA applies to subgroup 1 (see 4.3(d)).

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

(a) Country of origin.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 4. (See MIL-M-38510 Appendix E.)

4. PRODUCT ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and Method 5005 of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1, 4.4.2, and 4.4.3).

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4.3 Screening. Screening shall be in accordance with Method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

(a) Burn-in test (Method 1015 of MIL-STD-883).
    (1) Test condition D or E, using the circuit shown on figure 5, or equivalent.
    (2) T_A = 125°C minimum.
(b) Reverse bias burn-in and interim electrical test in accordance with 3.1.12 of Method 5004 of MIL-STD-883 may be omitted.
(c) Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
(d) Percent defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with Method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of Method 5005 of MIL-STD-883 and as follows:

(a) Tests shall be as specified in table II.
(b) Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of Method 5005 of MIL-STD-883.

4.4.3 Groups C and D inspections. Groups C and D inspections shall be in accordance with tables III and IV of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Subgroups 3 and 4 shall be added to the group C inspection requirements for class C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
(c) Operating life-test (Method 1005 of MIL-STD-883) conditions, or equivalent:
    (1) Test condition D or E, using the circuit shown on figure 5, or equivalent.
    (2) T_A = 125°C, minimum.
    (3) Test duration: 1,000 hours, except as permitted by Appendix B of MIL-M-38510.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

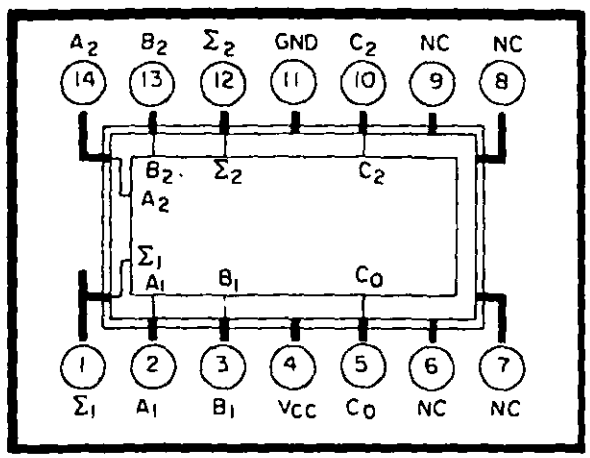
4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced to room temperature within 30 minutes after removal of the test condition.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with MIL-M-38510, except that the rough handling test shall not apply.

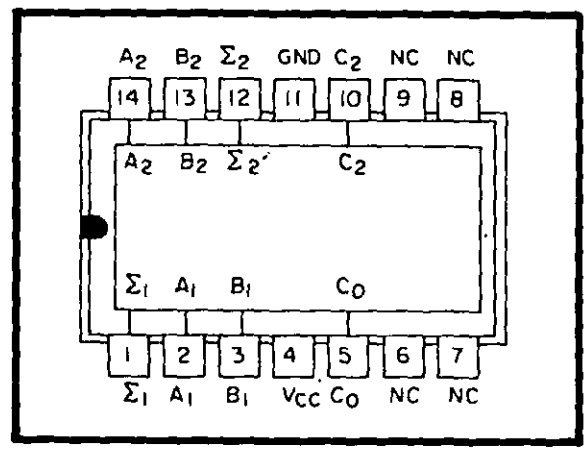
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Device type 01

Case A, B and D

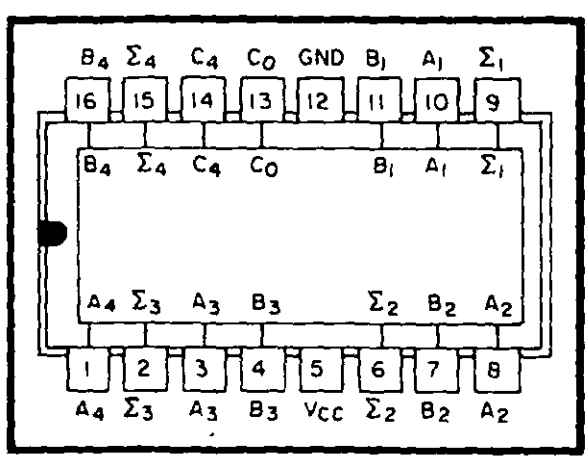


Case C



Device type 02

Case E and F



Device type 03

Case E and F

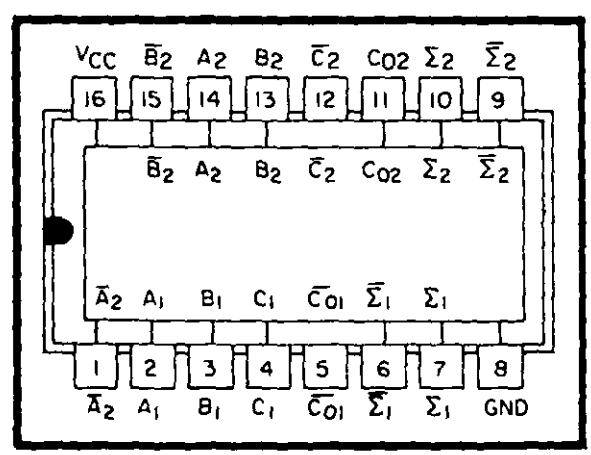
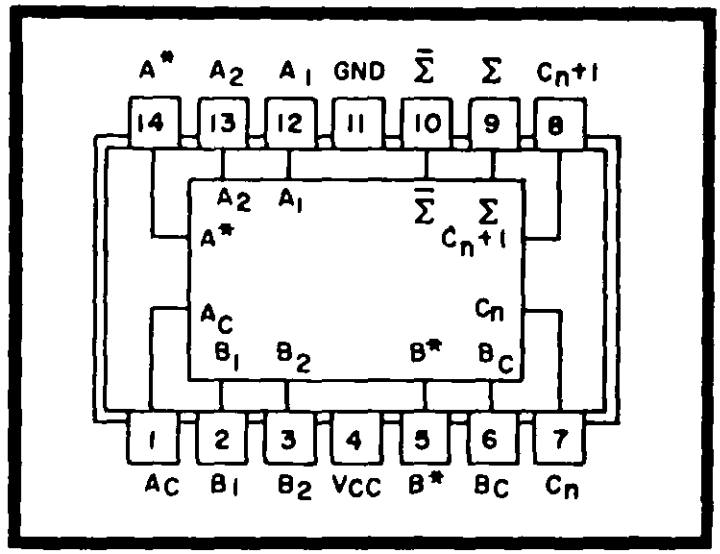


FIGURE 1. Terminal connections - top views.

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Device type 04

Cases A, B and D



Case C

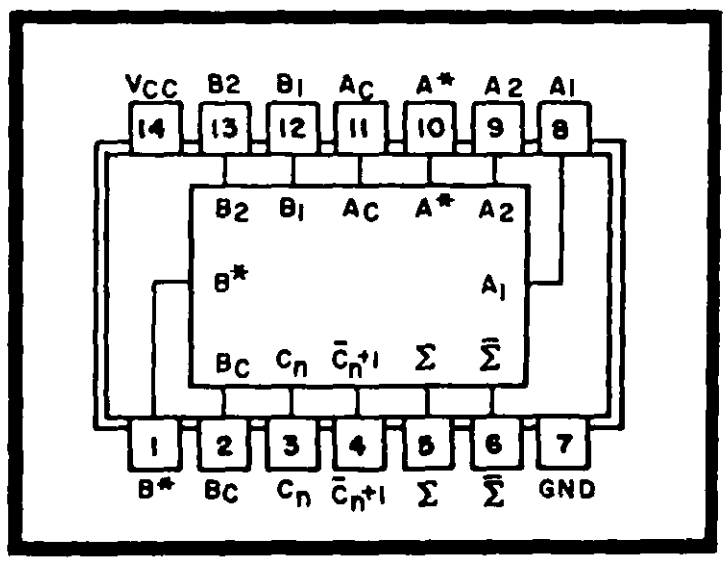


FIGURE 1. Terminal connections - top view - Continued.

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Device type 01

INPUT				OUTPUT					
				When C_0 = L			When C_0 = H		
A_1	B_1	A_2	B_2	Σ_1	Σ_2	C_2	Σ_1	Σ_2	C_2
L	L	L	L	L	L	L	H	L	L
H	L	L	L	H	L	L	L	H	L
L	H	L	L	H	L	L	L	H	L
H	H	L	L	L	H	L	H	H	L
L	L	H	L	L	H	L	H	H	L
H	L	H	L	H	H	L	L	L	H
L	H	H	L	H	H	L	L	L	H
H	H	H	L	L	L	H	H	L	H
L	L	L	H	L	H	L	H	H	L
H	L	L	H	H	H	L	L	L	H
L	H	L	H	H	H	L	L	L	H
H	H	L	H	L	L	H	H	L	H
L	L	H	H	L	L	H	H	L	H
H	L	H	H	H	L	H	L	H	H
L	H	H	H	H	L	H	L	H	H
H	H	H	H	L	H	H	H	H	H

Device type 02

INPUT				OUTPUT					
				When C_0 = L / When C_2 = L			When C_0 = H / When C_2 = H		
A_1 / A_3	B_1 / B_3	A_2 / A_4	B_2 / B_4	Σ_1 / Σ_3	Σ_2 / Σ_4	C_2 / C_4	Σ_1 / Σ_3	Σ_2 / Σ_4	C_2 / C_4
L	L	L	L	L	L	L	H	L	L
H	L	L	L	H	L	L	L	H	L
L	H	L	L	H	L	L	L	H	L
H	H	L	L	L	H	L	H	H	L
L	L	H	L	L	H	L	H	H	L
H	L	H	L	H	H	L	L	L	H
L	H	H	L	H	H	L	L	L	H
H	H	H	L	L	L	H	H	L	H
L	L	L	H	L	H	L	H	H	L
H	L	L	H	H	H	L	L	L	H
L	H	L	H	H	H	L	L	L	H
H	H	L	H	L	L	H	H	L	H
L	L	H	H	L	L	H	H	L	H
H	L	H	H	H	L	H	L	H	H
L	H	H	H	H	L	H	L	H	H
H	H	H	H	L	H	H	H	H	H

NOTE: Input conditions at A_1, A_2, B_1, B_2, and C_0 are used to determine outputs Σ_1 and Σ_2, and the value of the internal carry C_2. The values at C_2, A_3, B_3, A_4, and B_4, are then used to determine outputs Σ_3, Σ_4, and C_4.

FIGURE 2. Truth tables.

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Device type 03

Adder 2							
Inputs					Outputs		
$\overline{C}_2$	B_2	A_2	$\overline{B}_2$	$\overline{A}_2$	C_{O2}	$\Sigma 2$	$\overline{\Sigma 2}$
L	L	L	L	L	H	H	L
L	L	L	L	H	H	L	H
L	L	L	H	L	H	L	H
L	L	L	H	H	L	H	L
L	L	H	L	L	H	H	L
L	L	H	L	H	H	H	L
L	L	H	H	L	H	L	H
L	L	H	H	H	H	L	H
L	H	L	L	L	H	H	L
L	H	L	L	H	H	L	H
L	H	L	H	L	H	H	L
L	H	L	H	H	H	L	H
L	H	H	L	L	H	H	L
L	H	H	L	H	H	H	L
L	H	H	H	L	H	H	L
L	H	H	H	H	H	H	L
H	L	L	L	L	H	L	H
H	L	L	L	H	L	H	L
H	L	L	H	L	L	H	L
H	L	L	H	H	L	L	H
H	L	H	L	L	H	L	H
H	L	H	L	H	H	L	H
H	L	H	H	L	L	H	L
H	L	H	H	H	L	H	L
H	H	L	L	L	H	L	H
H	H	L	L	H	L	H	L
H	H	L	H	L	H	L	H
H	H	L	H	H	L	H	L
H	H	H	L	L	H	L	H
H	H	H	L	H	H	L	H
H	H	H	H	L	H	L	H
H	H	H	H	H	L	L	H

Adder 1					
Inputs			Outputs		
C_1	B_1	A_1	$\overline{C}_{O1}$	$\overline{\Sigma 1}$	$\Sigma 1$
L	L	L	H	H	L
L	L	H	H	L	H
L	H	L	H	L	H
L	H	H	L	H	L
H	L	L	H	L	H
H	L	H	L	H	L
H	H	L	L	H	L
H	H	H	L	L	H

Device type 04

Inputs			Outputs		
C_n	B	A	$\overline{C}_{n+1}$	$\overline{\Sigma}$	Σ
L	L	L	H	H	L
L	L	H	H	L	H
L	H	L	H	L	H
L	H	H	L	H	L
H	L	L	H	L	H
H	L	H	L	H	L
H	H	L	L	H	L
H	H	H	L	L	H

NOTES:
1. $A = \overline{A}_C + \overline{A}^* + A1 \cdot A2$, $B = \overline{B}_C + \overline{B}^* + B1 \cdot B2$.
2. When A* is used as an input, A1 or A2 must be low.
   When B* is used as an input, B1 or B2 must be low.
3. When A1 and A2 or B1 and B2 are used as inputs, A* or B*, respectively, must be open or used to perform dot-AND logic.

FIGURE 2. Truth tables - Continued.

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Device type 01, Circuit A

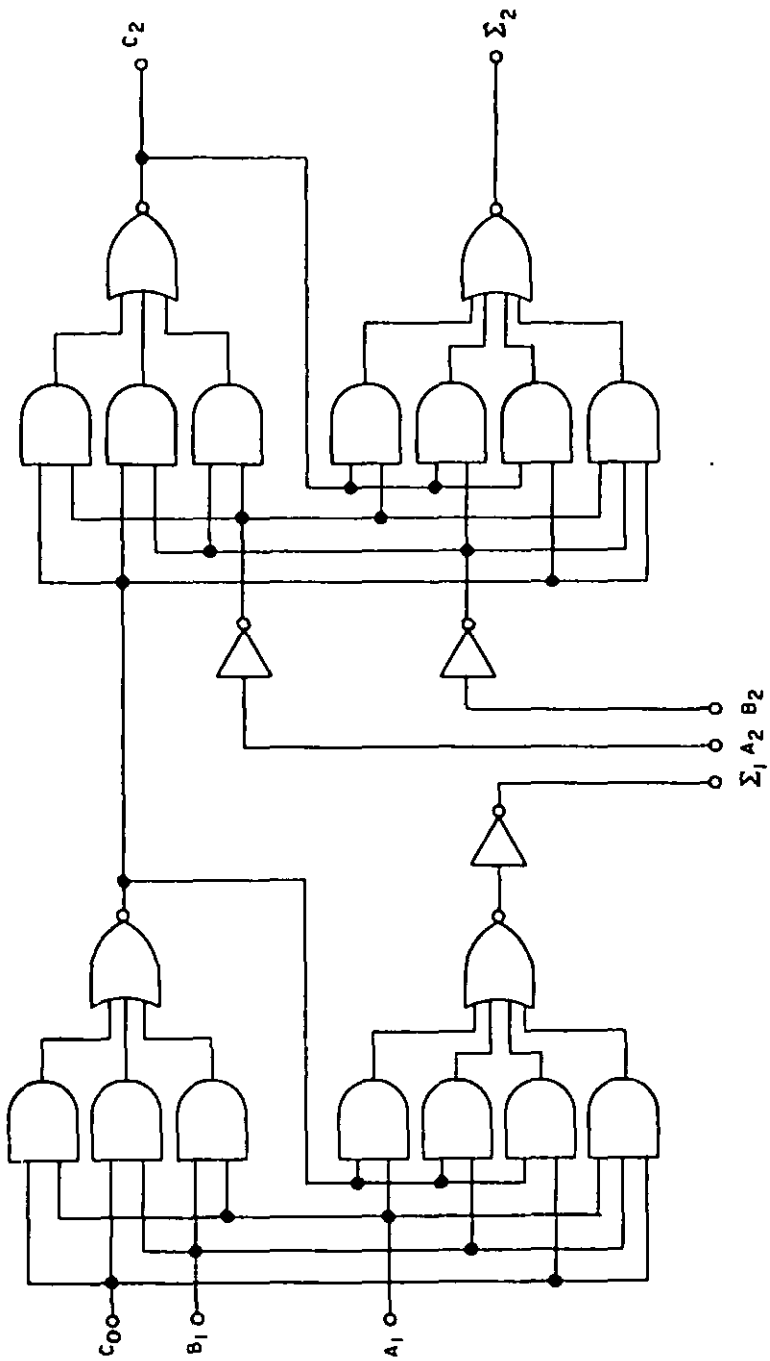


FIGURE 3. Logic diagrams.

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Device type 01, Circuit B

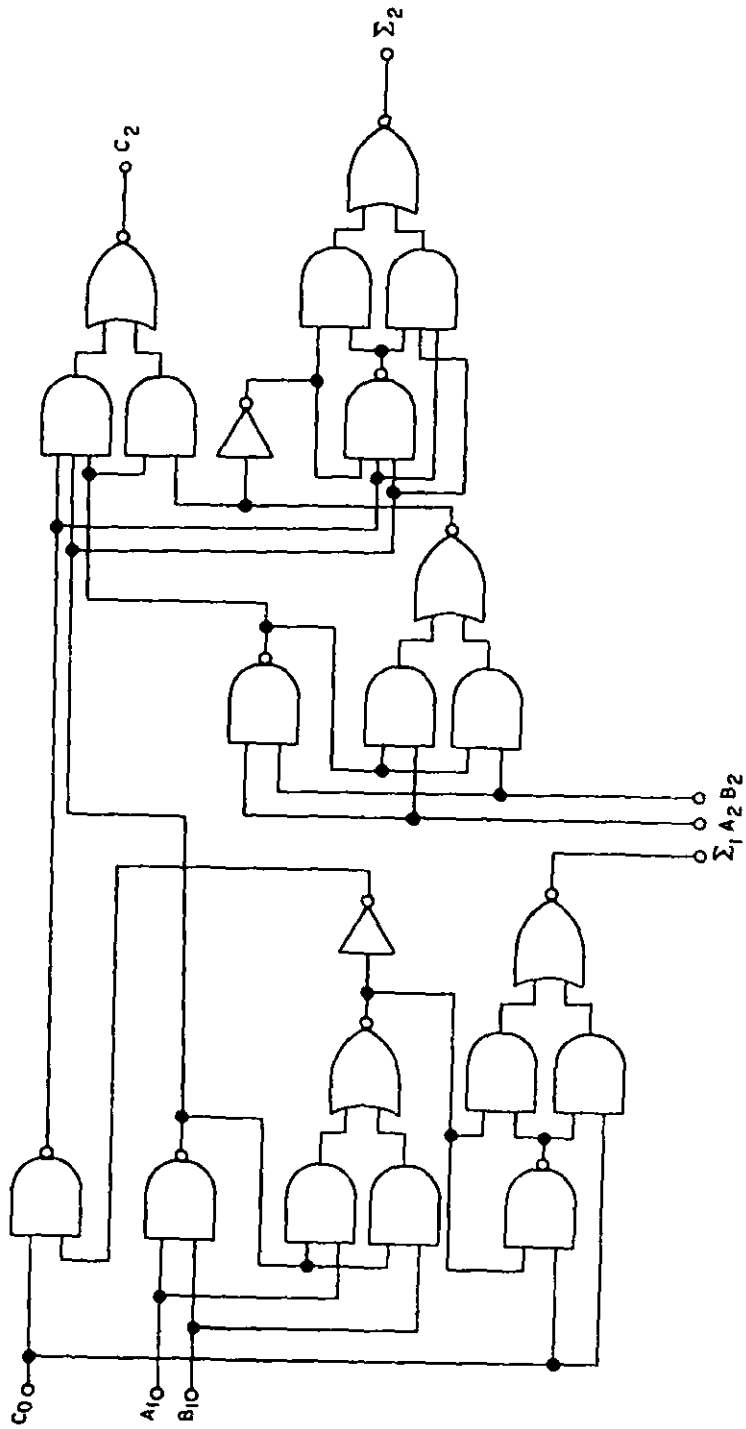


FIGURE 3. Logic diagrams - Continued.

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Device type 02, Circuit C, E, G and H

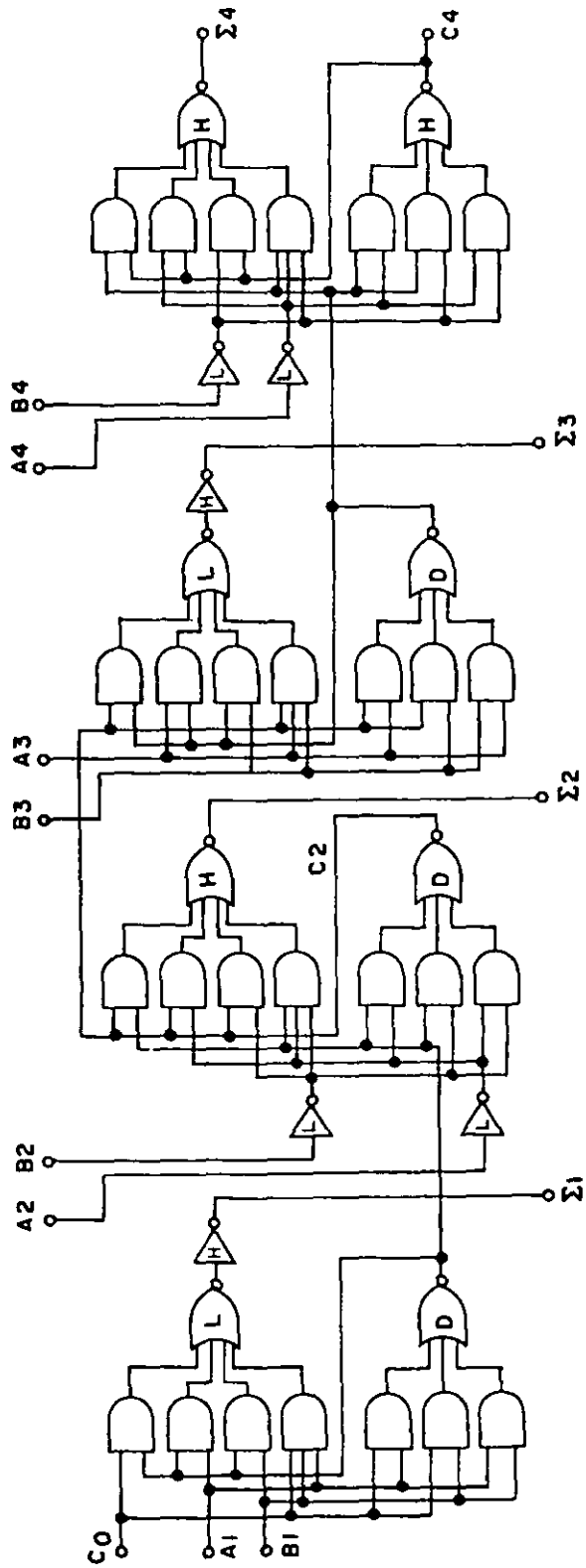


FIGURE 3. Logic diagrams - Continued.

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Device type 02, Circuit D

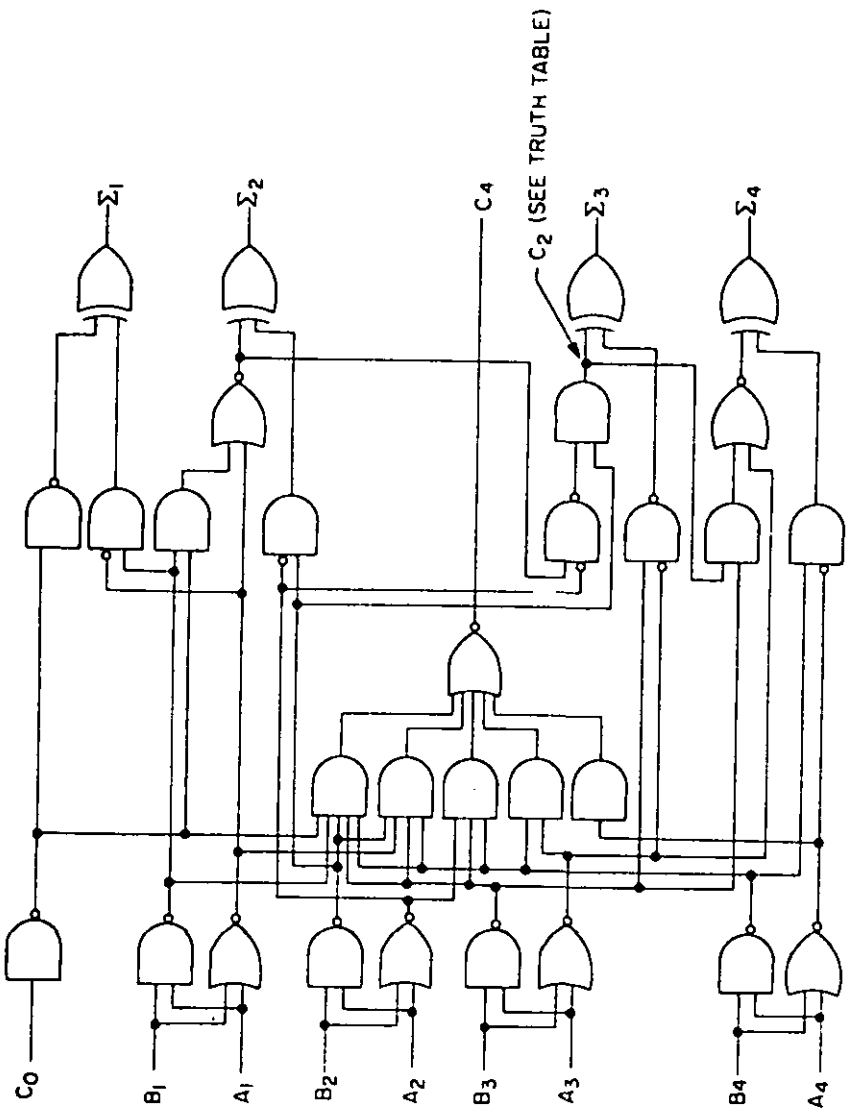


FIGURE 3. Logic diagrams - Continued.

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Device type 02, Circuit F

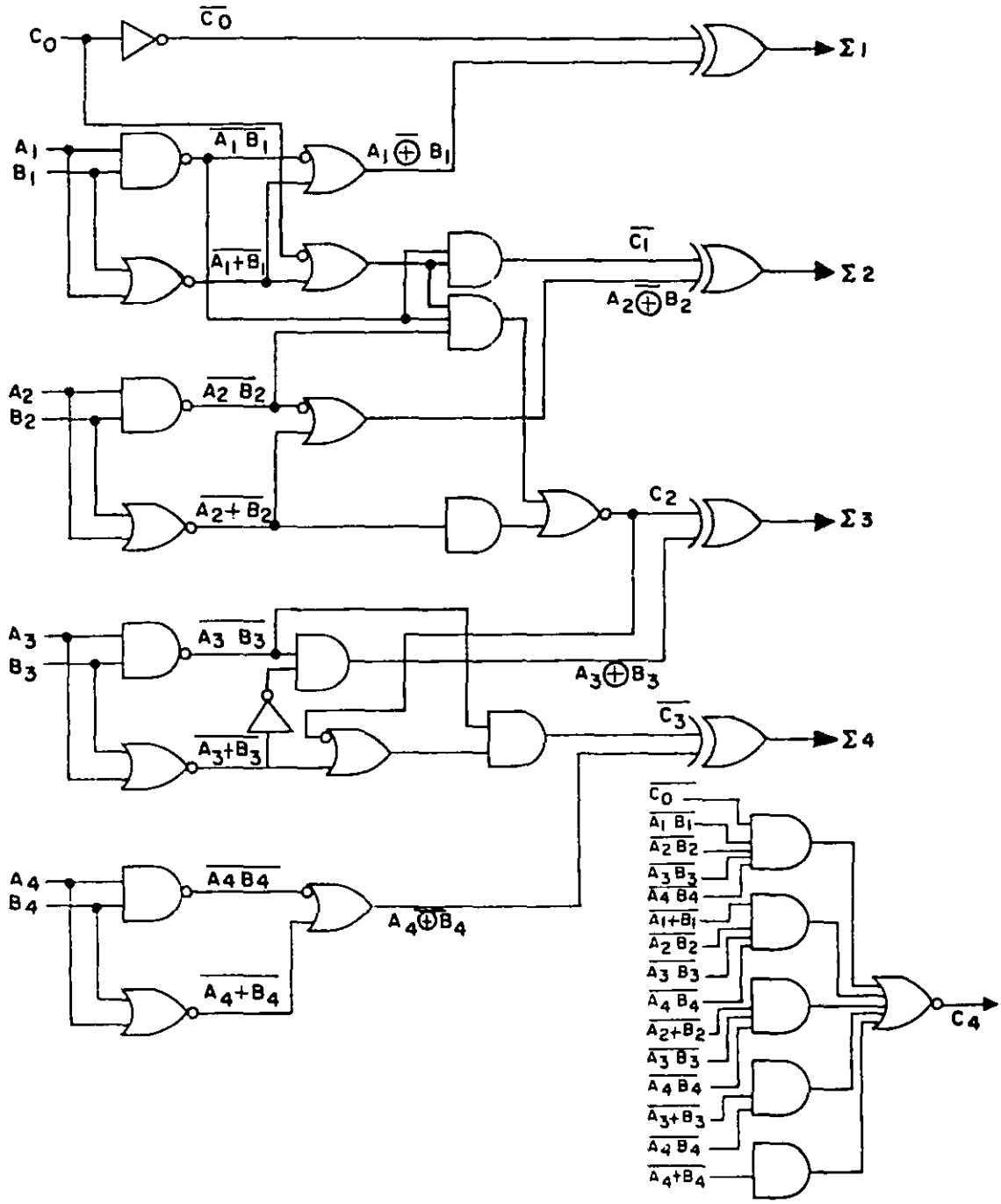


FIGURE 3. Logic diagrams - Continued.

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*Device type 03, circuit J.

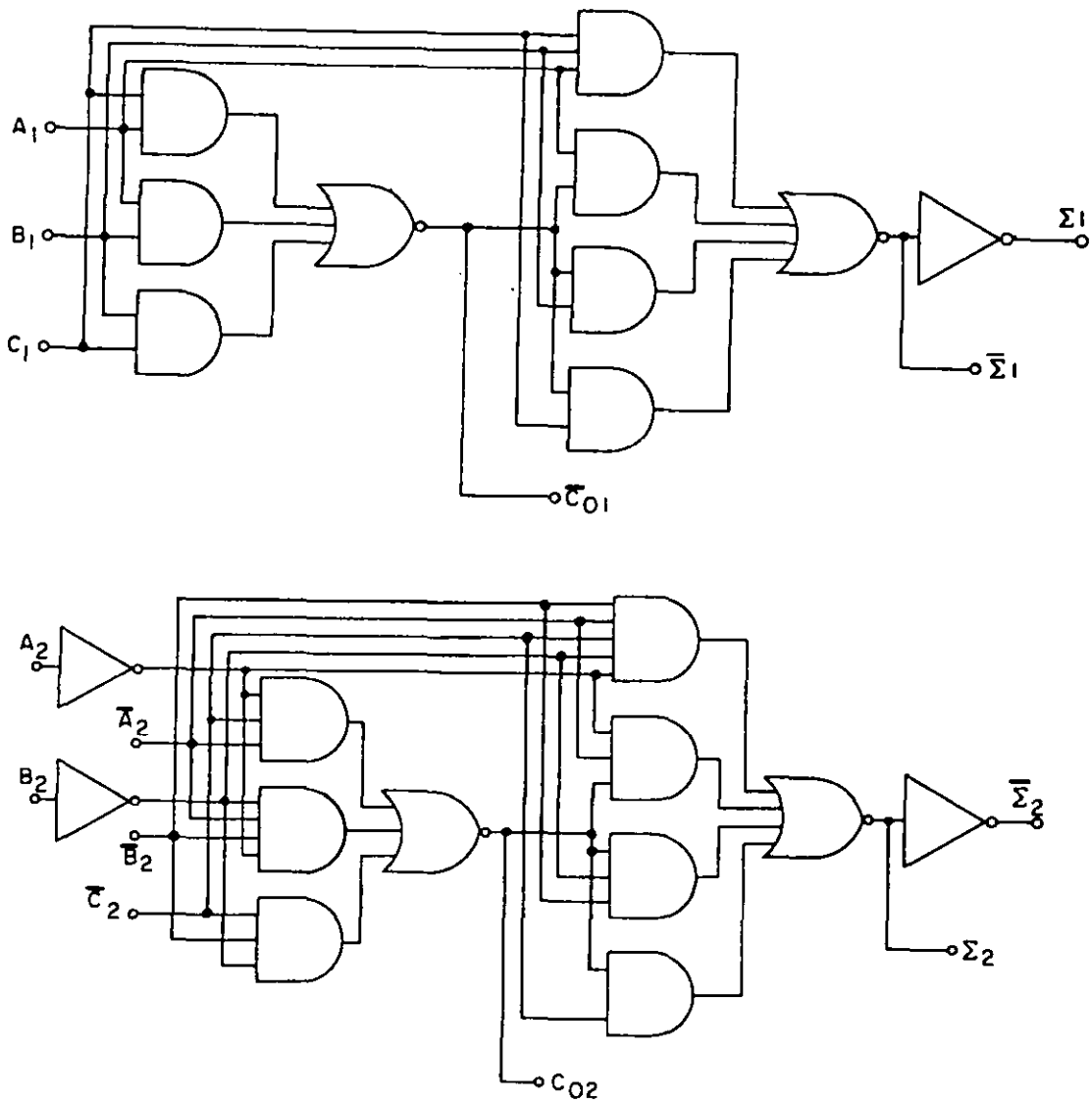


FIGURE 3. Logic diagrams - Continued.

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Device type 04

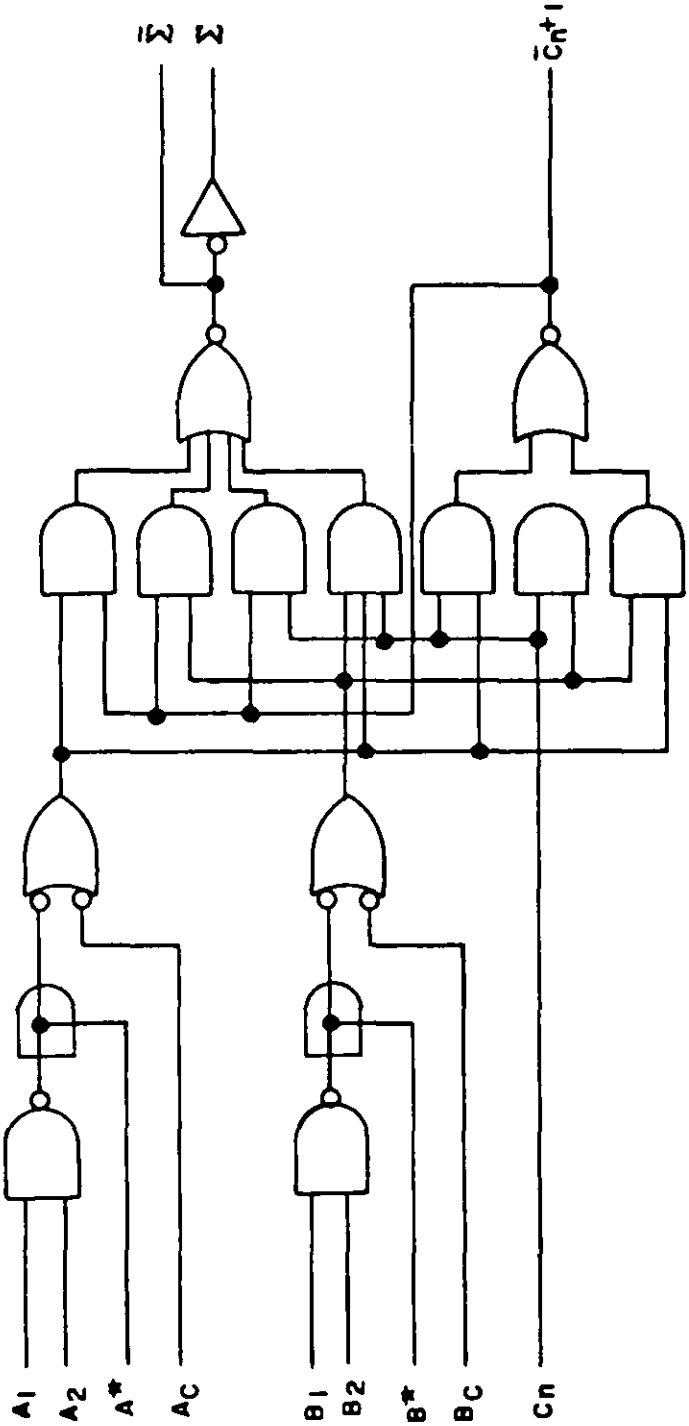
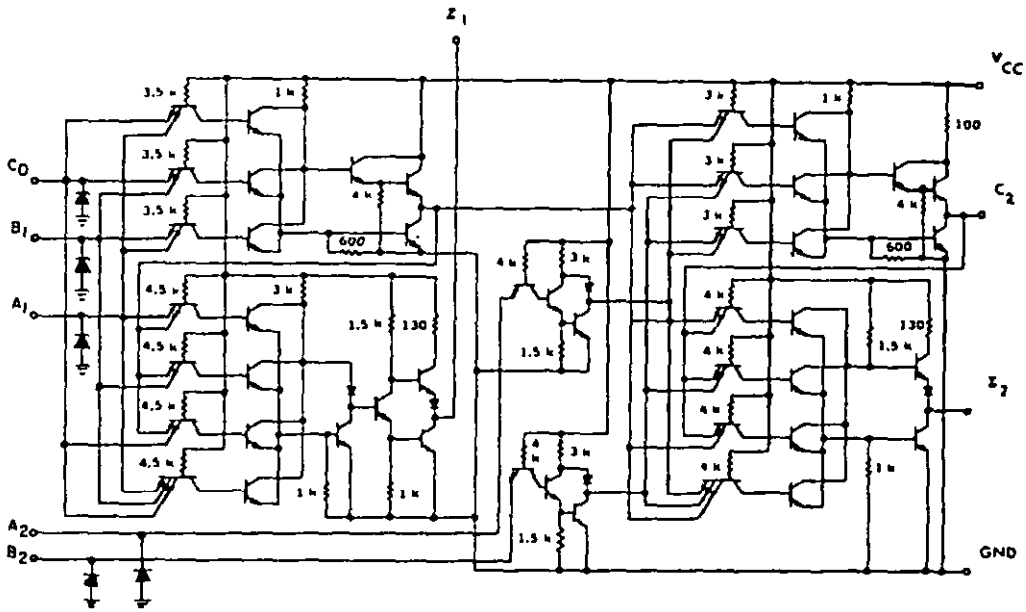


FIGURE 3. Logic diagrams - Continued.

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Device type 01, Circuit A



NOTES:
1. Circuits A and B correspond to logic diagrams A and B respectively. These are the only acceptable circuits and logic diagrams for device type 01.
2. Circuits C, G, and H correspond to logic diagram C and circuits D, E, and F correspond to logic diagrams D, E, and F, respectively. These are the only acceptable circuits and logic diagrams for type 02.
3. Circuit J corresponds to logic diagram J. This is the only acceptable circuit and logic diagram for type 03.
4. Component values shown are nominal and resistor values are in ohms.

FIGURE 4. Schematic circuits.

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Device type 01, Circuit B

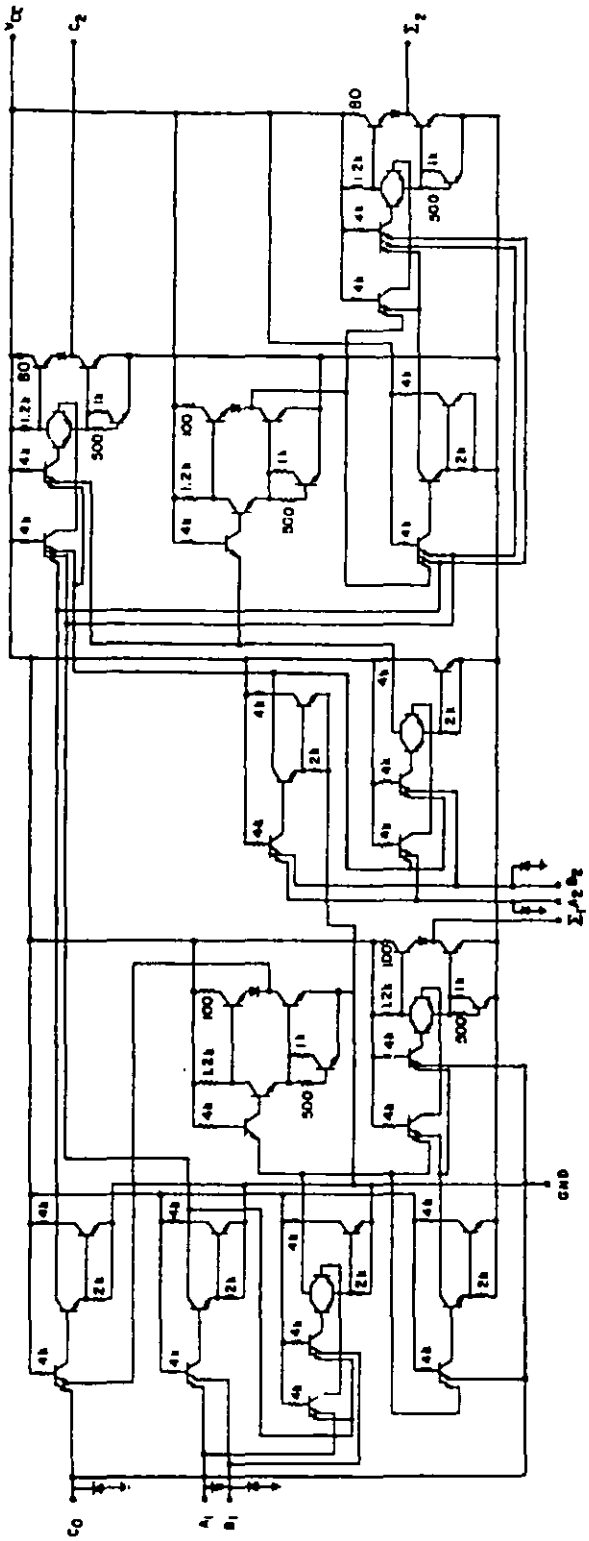


FIGURE 4. Schematic circuits - Continued.

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Device type 02. Circuit C

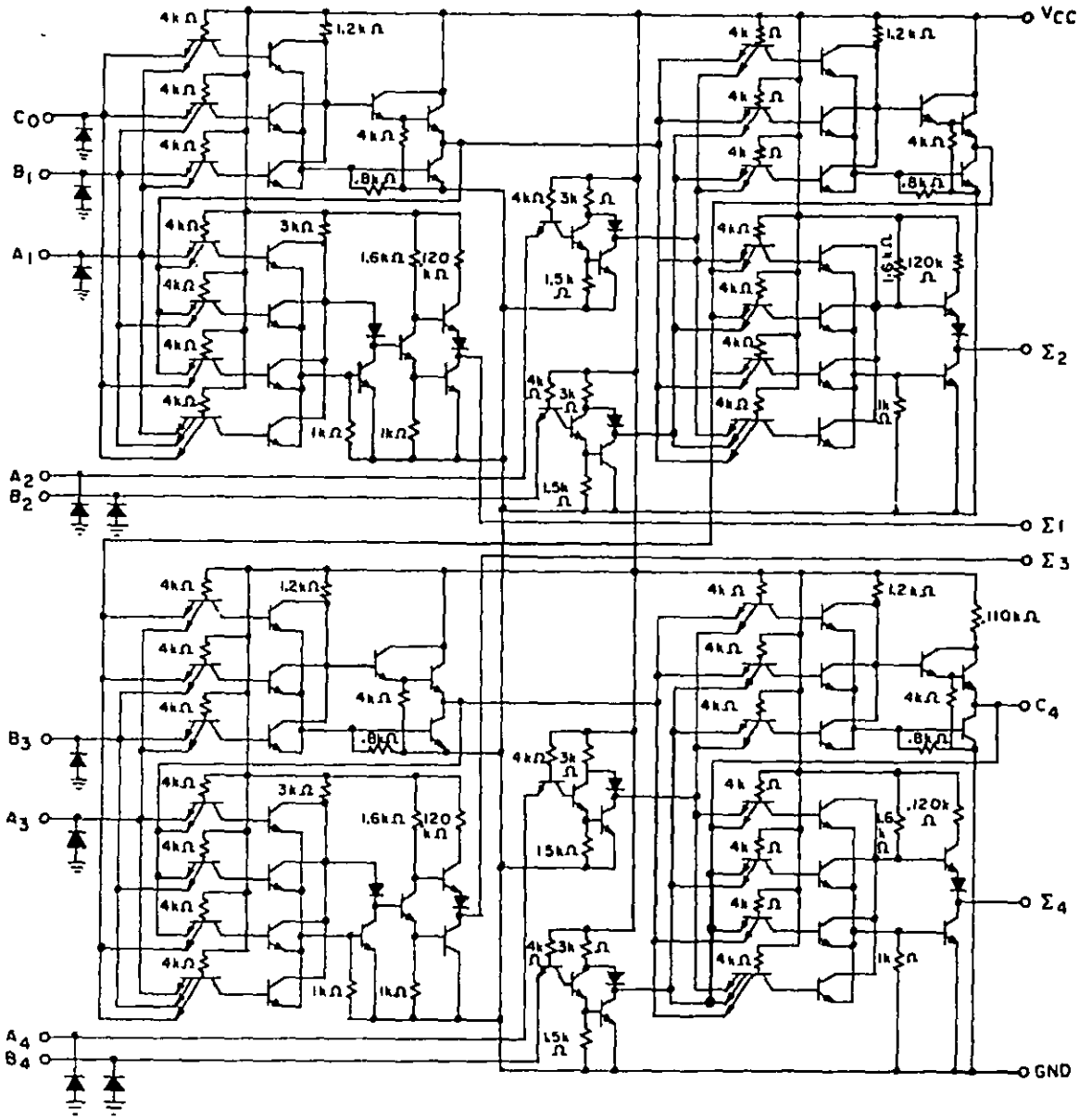


FIGURE 4. Schematic circuits - Continued.

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Device type 02, Circuit D

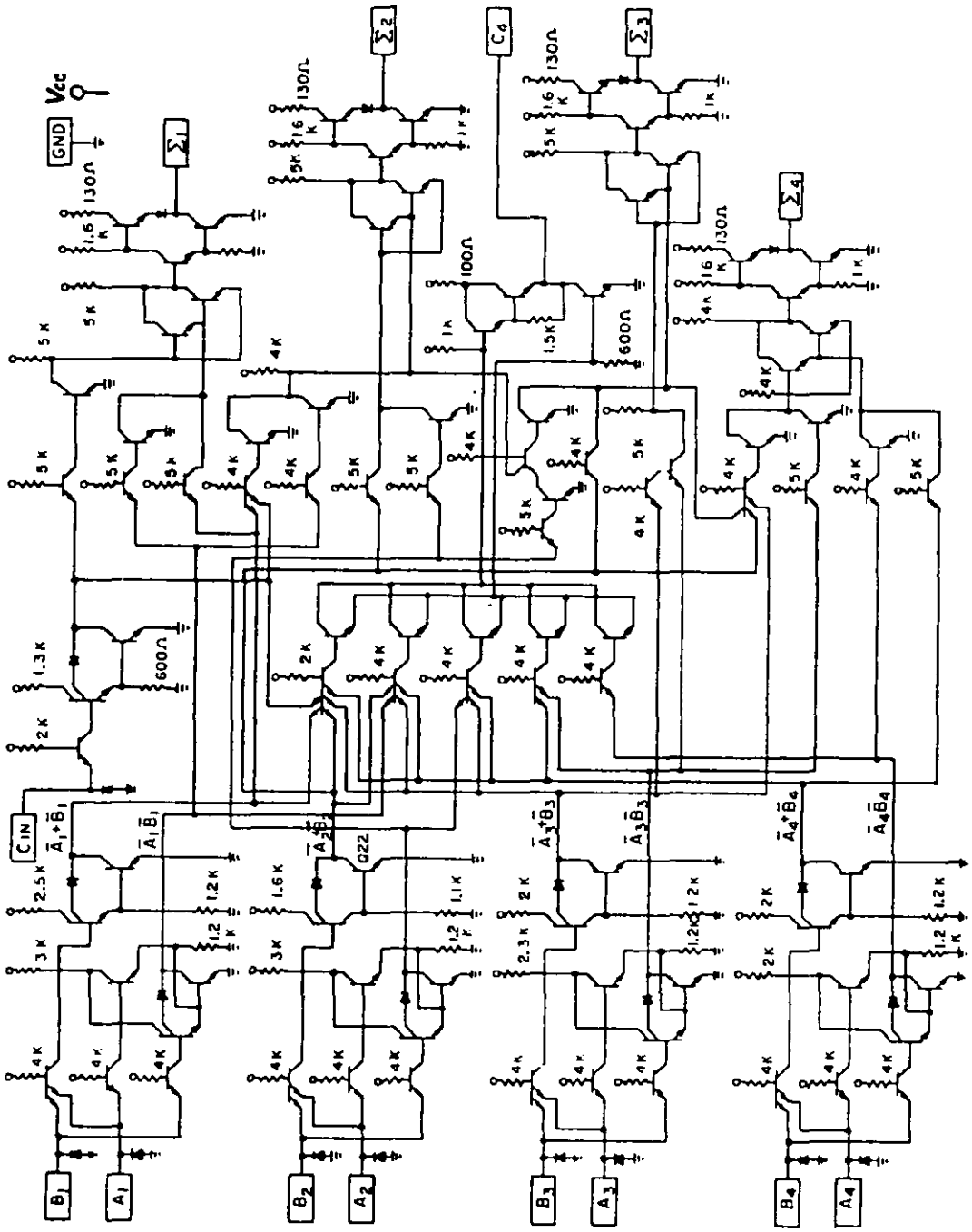


FIGURE 4. Schematic circuits - Continued.

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Device type 02, Circuit E.

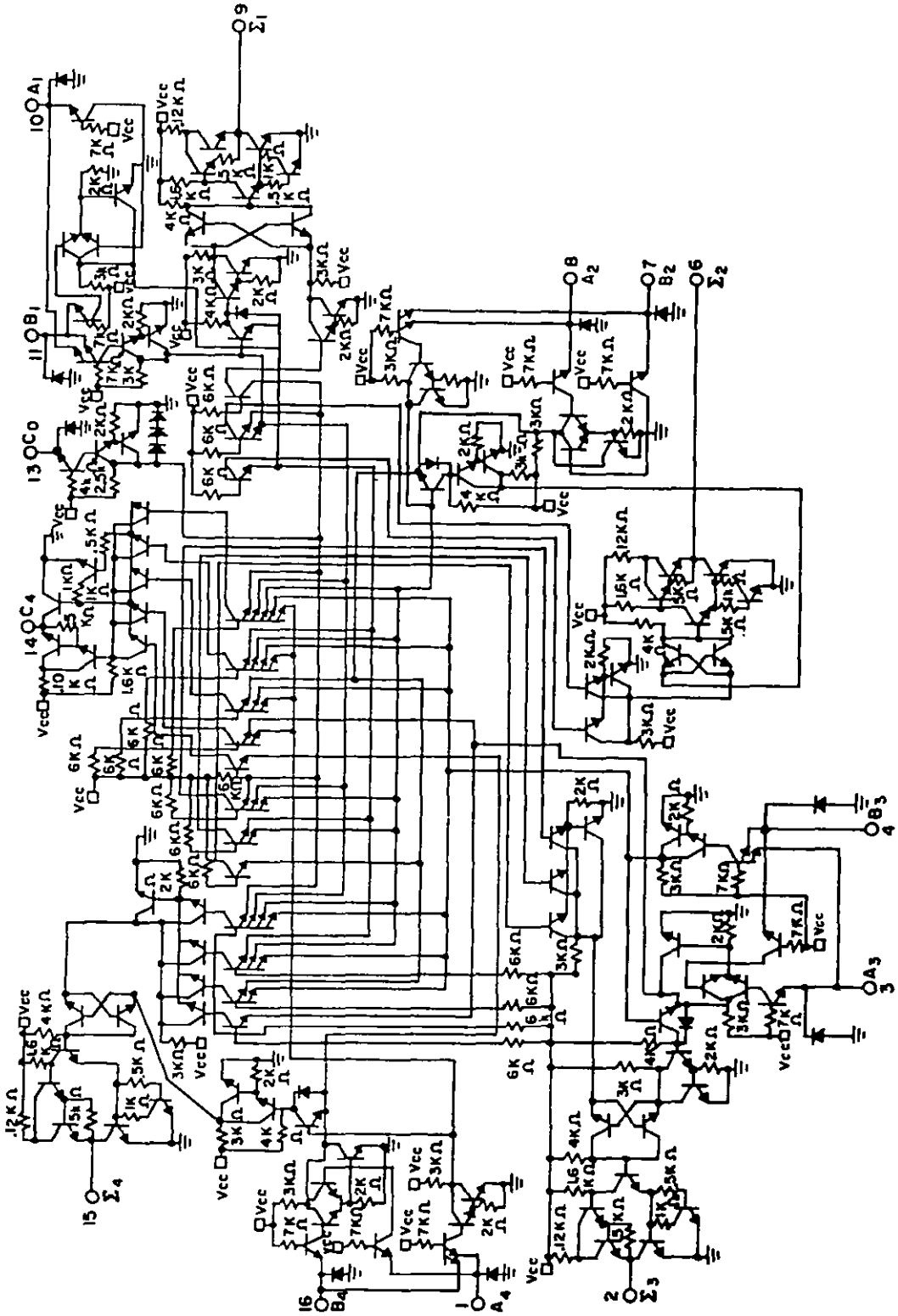


FIGURE 4. Schematic circuits - Continued.

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Device type 02, Circuit F

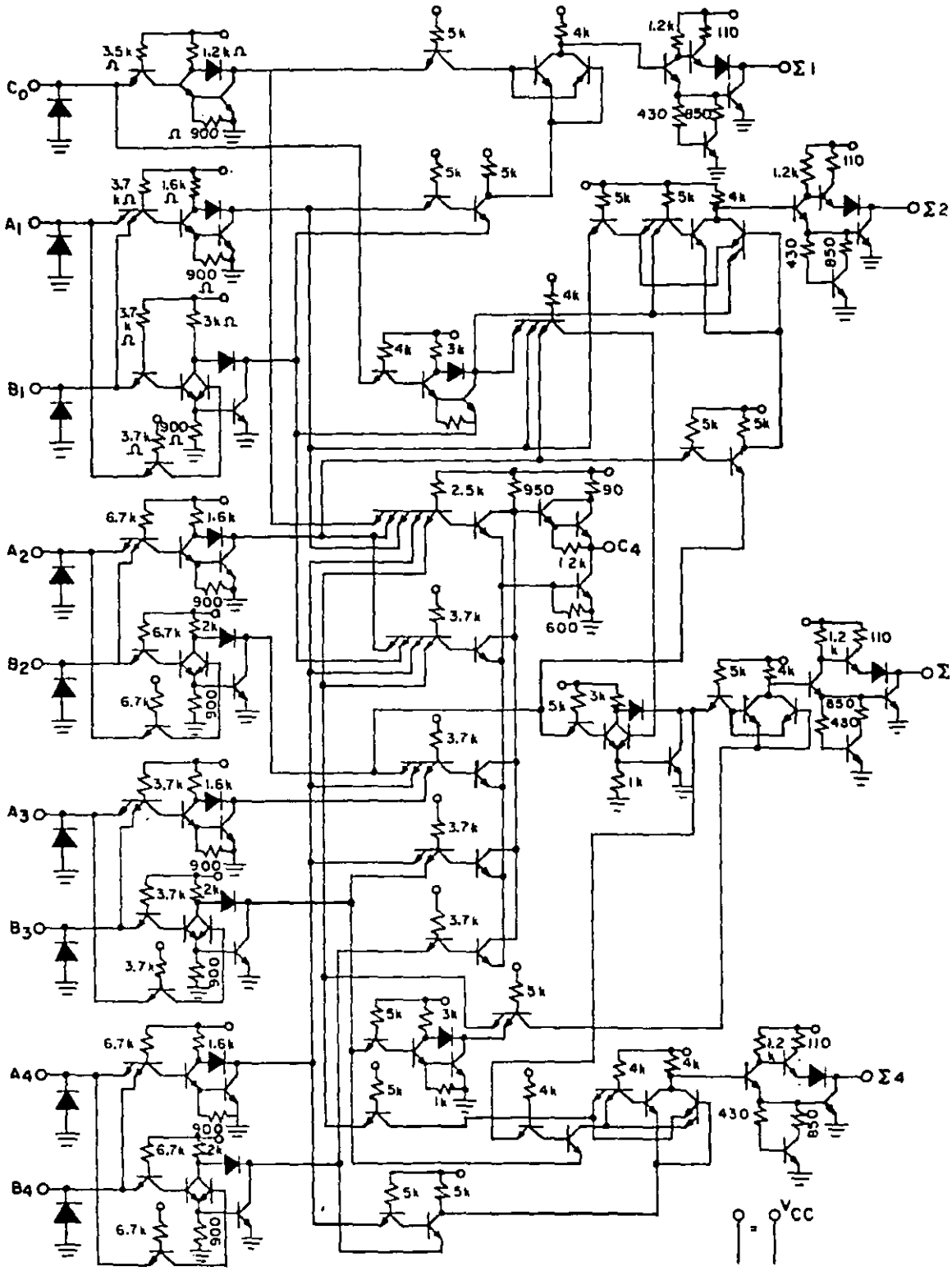


FIGURE 4. Schematic circuits - Continued.

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Device type 02, Circuit G

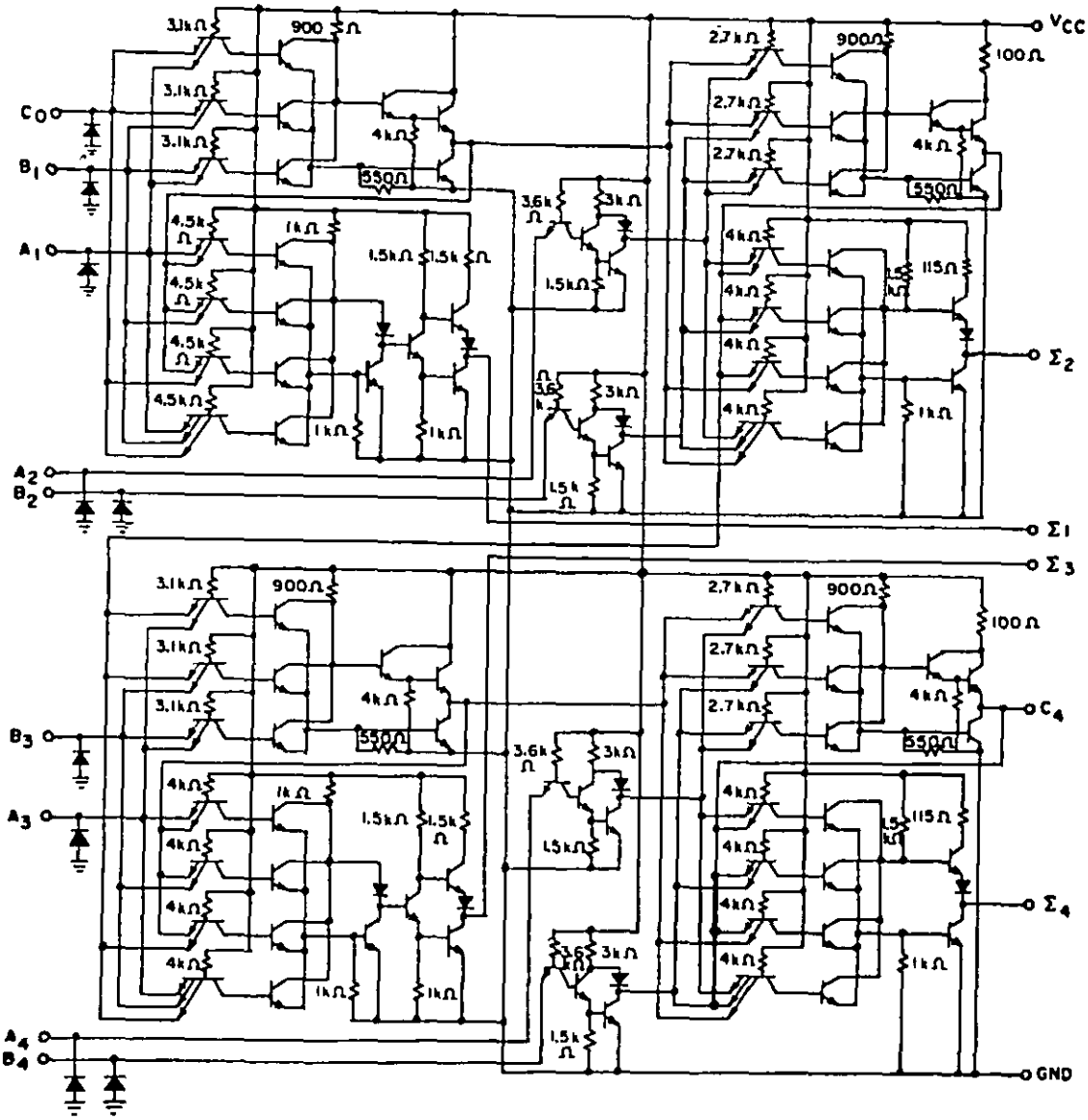


FIGURE 4. *Schematic circuits - Continued.*

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Device type 02, circuit H

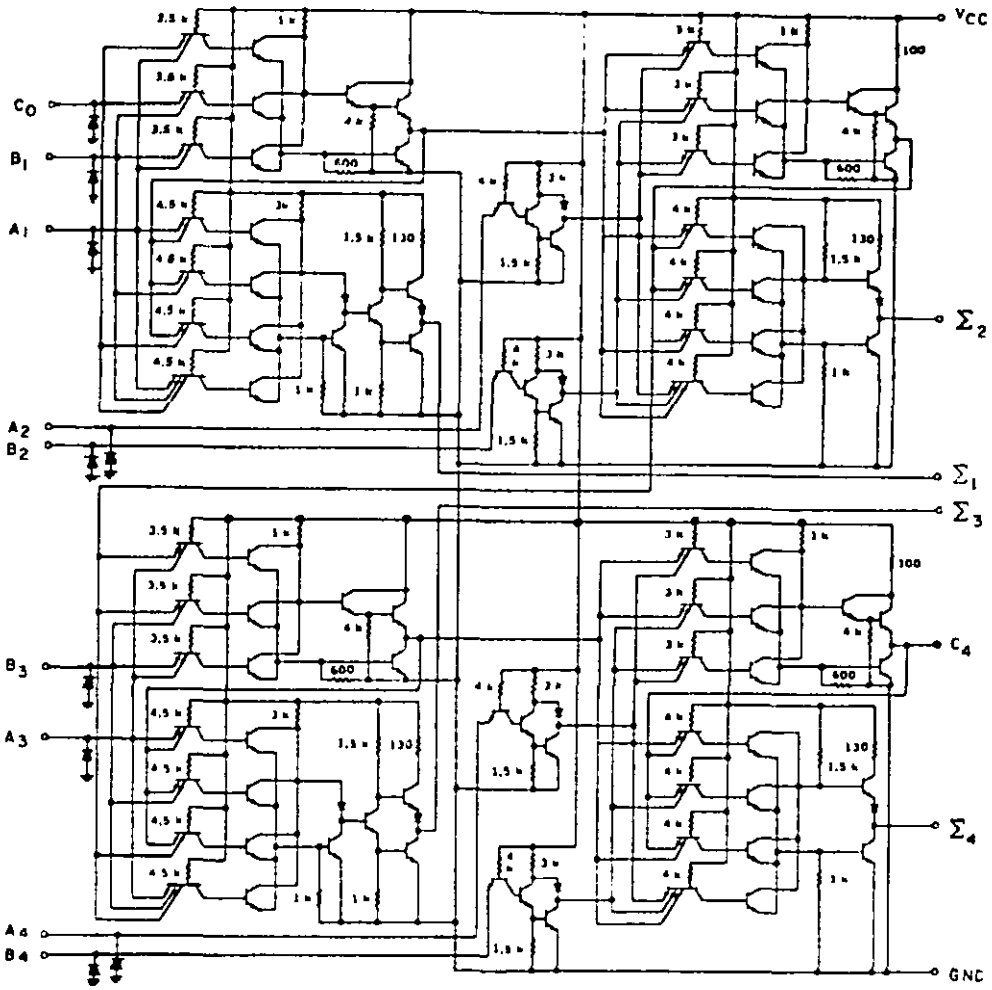
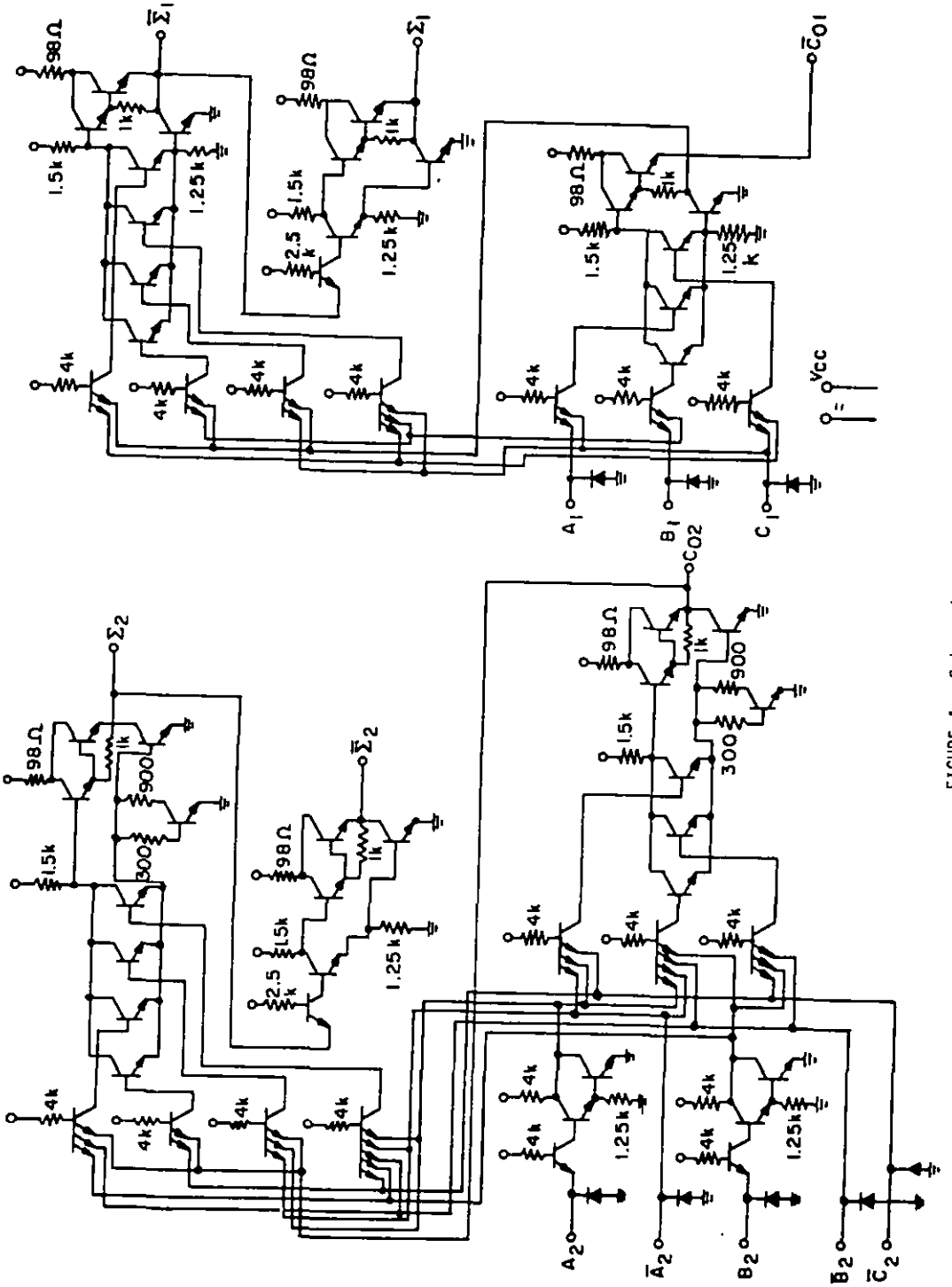


FIGURE 4. Schematic circuits - Continued.

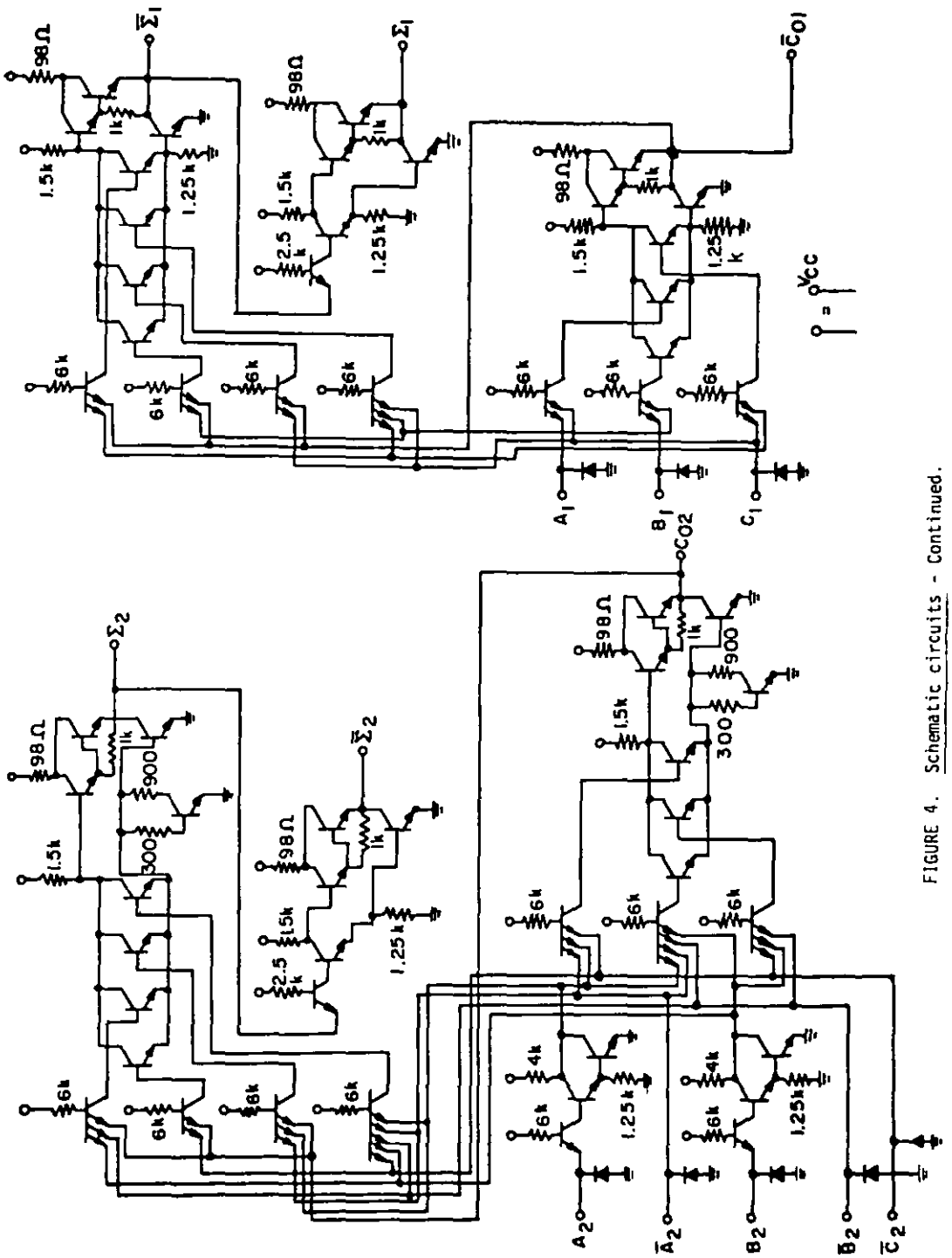
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*Device type 03, Circuit J.

FIGURE 4. Schematic circuits - Continued.

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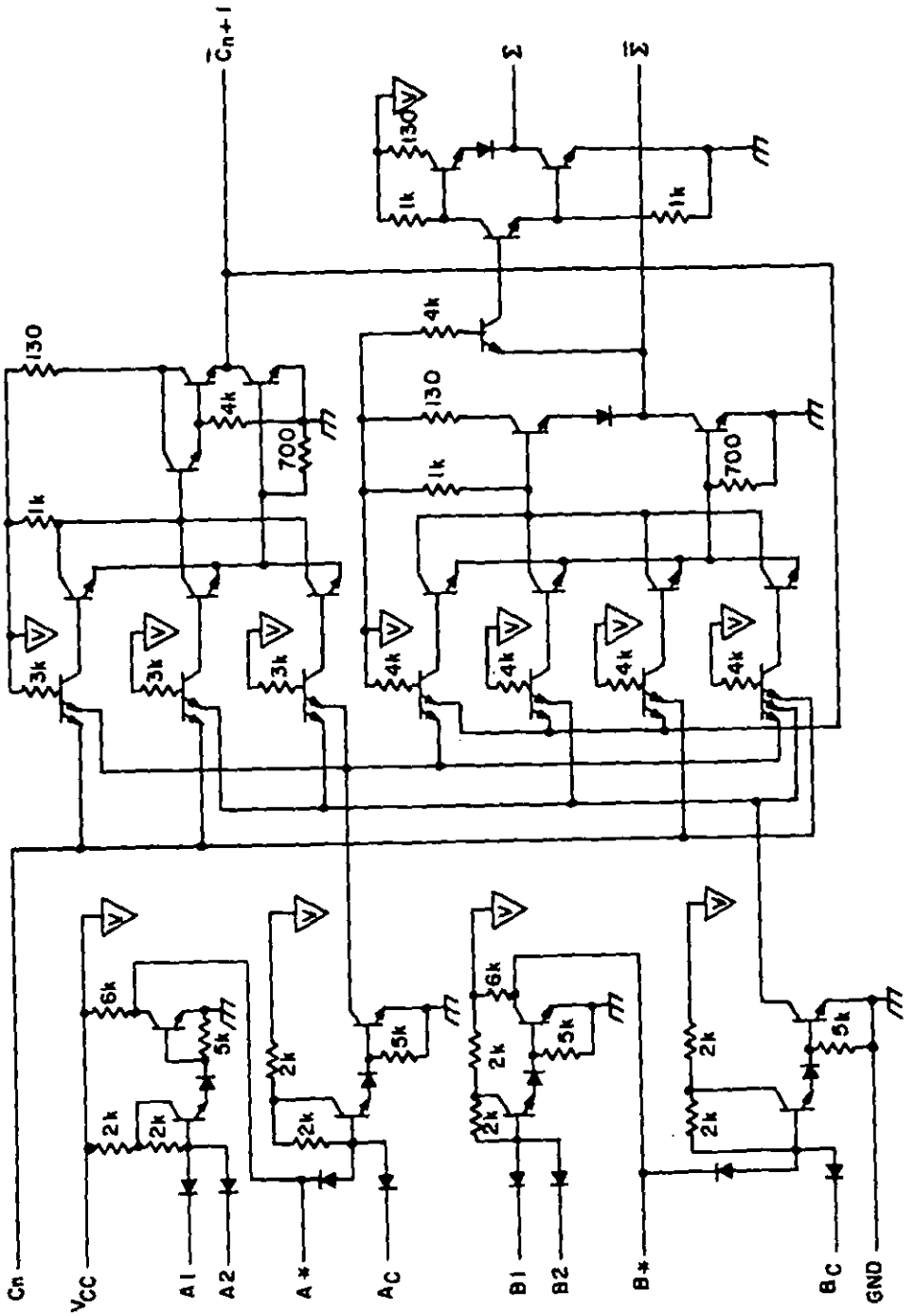


*Device type 03, Circuit K.

FIGURE 4. Schematic circuits - Continued.

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Device type 04

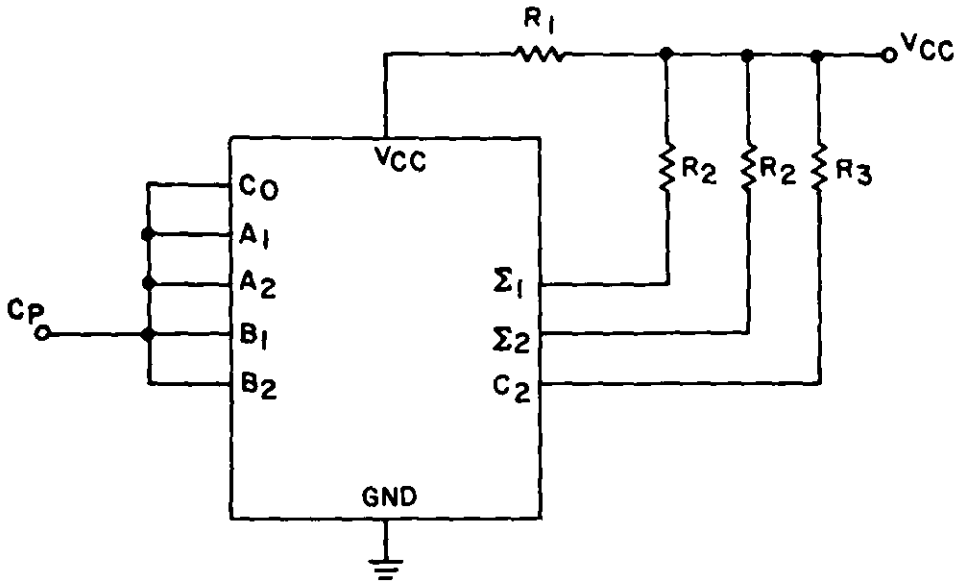


Resistor values shown are nominal and in ohms.

FIGURE 4. Schematic circuits - Continued.

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Device type 01

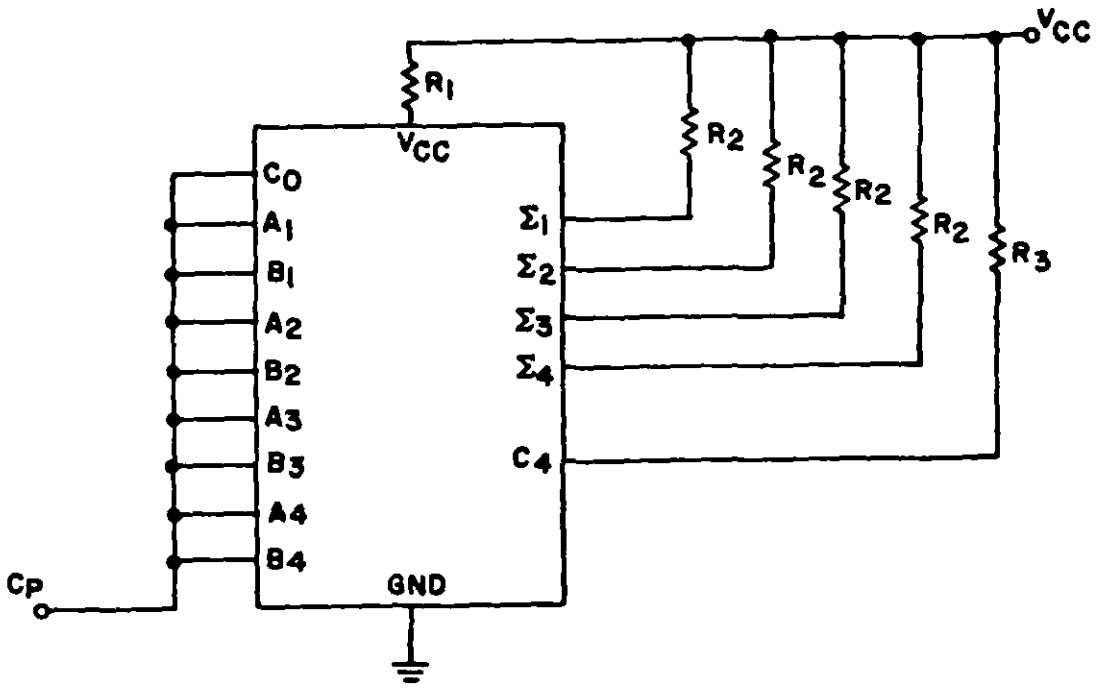


NOTES:
1. V_{CC} and R_1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R_2 = 270Ω ±5%.
3. R_3 = 560Ω ±5%.
4. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 5. Burn-in and life test circuits.

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Device type 02

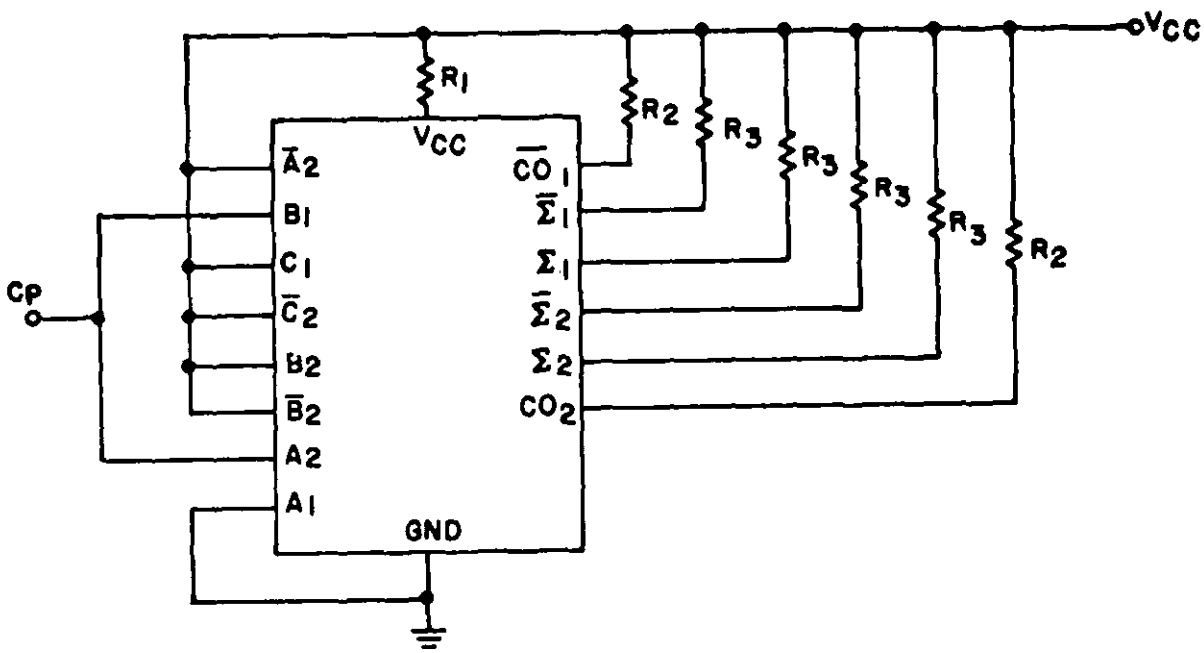


NOTES:
1. V_{CC} and R_1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R_2 = 270Ω ±5%.
3. R_3 = 560Ω ±5%.
4. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 5. Burn-in and life test circuits - Continued.

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Device type 03

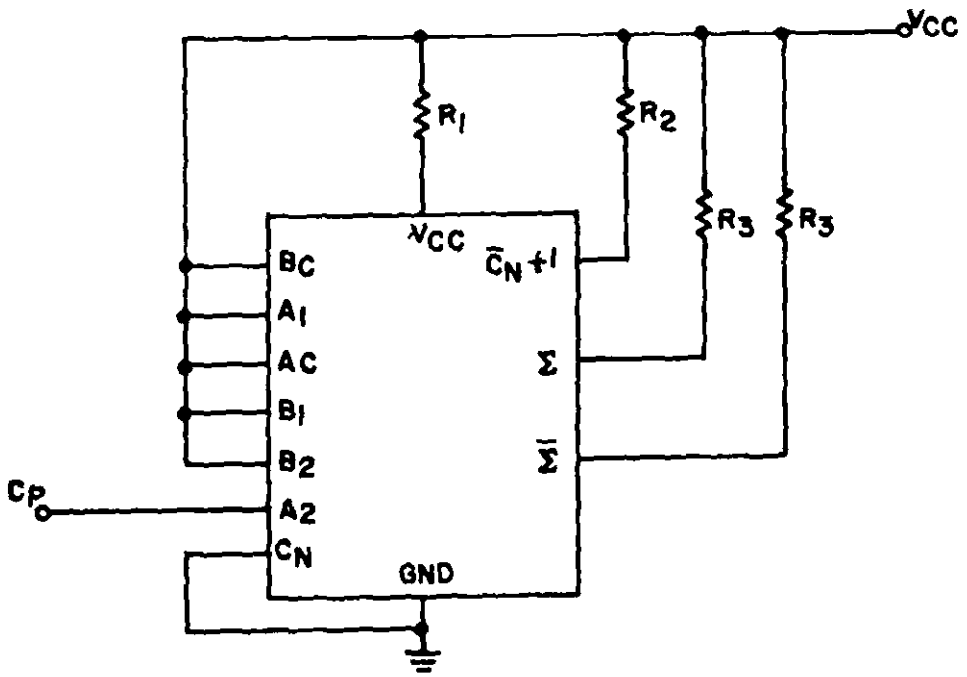


NOTES:
1. V_{CC} and R_1 are such that the minimum voltage at the device terminal is 5.0 volts.
2. R_2 = 390Ω ±5%.
3. R_3 = 270Ω ±5%.
4. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 5. Burn-in and life test circuits - Continued.

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Check the source to verify that this is the current version before use.

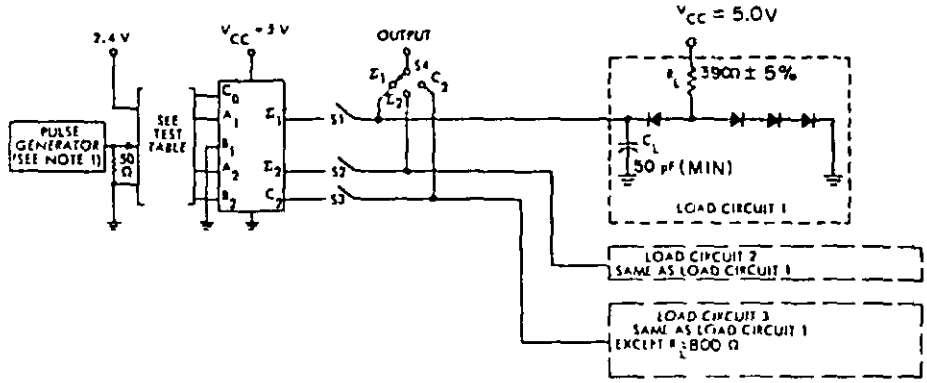
Device type 04

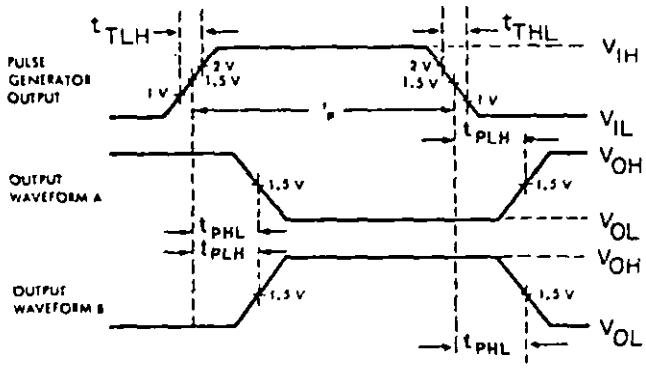


NOTES:
1. R_1 and V_{CC} are such that the minimum voltage at the device terminal is 5.0 volts.
2. R_2 = 560Ω ±5%.
3. R_3 = 270Ω ±5%.
4. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.

FIGURE 5. Burn-in and life test circuits - Continued.

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SWITCHING TIMES TEST TABLE (SEE NOTE 7)

PARAMETER	APPLY PULSE GENERATOR OUTPUT TO	OUTPUT UNDER TEST (S4)	APPLY 2.4 V TO	APPLY GND TO	S1	S2	S3
t_{PLH_1} t_{PHL_1}	C_0	Σ_1 (WAVEFORM A)	A_1	A_2, B_1, B_2	CLOSED	OPEN	OPEN
t_{PLH_2} t_{PHL_2}	B_2	Σ_2 (WAVEFORM B)	None	A_1, B_1, A_2 and C_0	OPEN	CLOSED	OPEN
t_{PLH_3} t_{PHL_3}	C_0	Σ_2 (WAVEFORM A)	A_1, A_2	B_1, B_2	OPEN	CLOSED	CLOSED
t_{PLH_4} t_{PHL_4}	C_0	C_2 (WAVEFORM B)	A_1, A_2	B_1, B_2	OPEN	OPEN	CLOSED

NOTES:
1. The generator has the following characteristics: $V_{IH} \geq 2.4V$, $V_{IL} \leq 0.4V$, t_{TLH} = 8 to 15 ns, t_{THL} = 3 to 5 ns, PRR = 1 mHz, t_p = 200 ns, and $Z_{out} \approx 50\Omega$.
2. Perform test in accordance with test table.
3. Each output is tested separately.
4. Voltage values are with respect to network ground terminal.
5. C_L includes probe and jig capacitance.
6. All diodes are 1N3064, or equivalent.
7. Inputs and outputs not specified are open.

FIGURE 6. Switching time test circuit - device type 01.

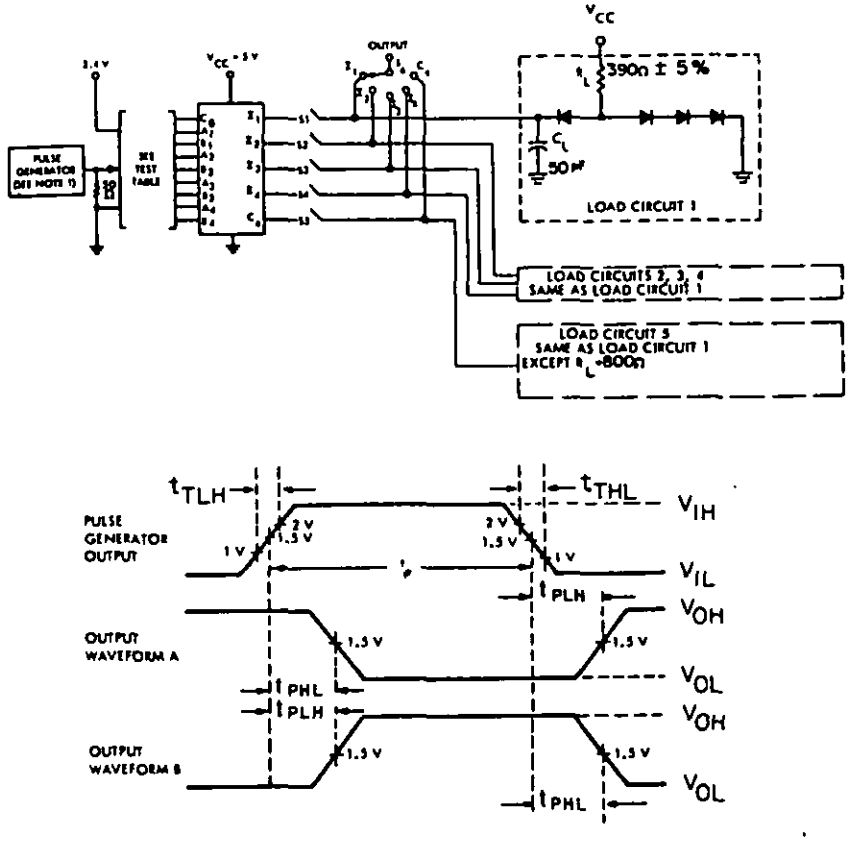
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:47Z
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TEST TABLE

PARAMETER	APPLY PULSE GENERATOR OUTPUT TO	OUTPUT UNDER TEST (S6)	APPLY 2.4 V TO	APPLY GND TO	S1	S2	S3	S4	S5
t_{PLH_1} t_{PHL_1}	C_0	Σ_1 (WAVEFORM A)	A_1	B_1, A_2, and B_2	CLOSED	OPEN	OPEN	OPEN	OPEN
t_{PLH_2} t_{PHL_2}	B_2	Σ_2 (WAVEFORM B)	None	A_1, B_1, A_2, and C_0	OPEN	CLOSED	OPEN	OPEN	OPEN
t_{PLH_3} t_{PHL_3}	C_0	Σ_3 (WAVEFORM A)	A_1 and A_2	B_1 and B_2	OPEN	CLOSED	OPEN	OPEN	OPEN
t_{PLH_4} t_{PHL_4}	C_0	Σ_3 (WAVEFORM A)	A_1, A_2, and A_3	B_1, B_2, and B_3	OPEN	OPEN	CLOSED	OPEN	OPEN
t_{PLH_5} t_{PHL_5}	C_0	Σ_4 (WAVEFORM A)	A_1, A_2, A_3, and A_4	B_1, B_2, B_3, and B_4	OPEN	OPEN	OPEN	CLOSED	CLOSED
t_{PLH_6} t_{PHL_6}	C_0	C_4 (WAVEFORM B)	A_1, A_2, A3, and A_4	B_1, B_2, B_3, and B_4	OPEN	OPEN	OPEN	OPEN	CLOSED
t_{PLH_7} t_{PHL_7}	A_2	Σ_2 (WAVEFORM B)	None	A_1, B_1, B_2, and C_0	OPEN	CLOSED	OPEN	OPEN	OPEN
t_{PLH_8} t_{PHL_8}	A_4	Σ_4 (WAVEFORM B)	None	A_3, B_3, and B_4	OPEN	OPEN	OPEN	CLOSED	OPEN
t_{PLH_9} t_{PHL_9}	B_4	Σ_4 (WAVEFORM B)	None	A_3, B_3, and A_4	OPEN	OPEN	OPEN	CLOSED	OPEN

FIGURE 7. Switching time test circuit - device type 02.

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NOTES:
1. The generator has the following characteristics: $V_{IH} \geq 2.4V$, $V_{IL} \leq 0.4V$, t_{TLH} = 8 to 15 ns, t_{THL} = 3 to 5 ns, PRR = 1 mHz, t_p = 200 ns, and $Z_{out} \approx 50\Omega$.
2. Perform test in accordance with test table. (See sheet 2 of this figure).
3. Each output is tested separately.
4. Voltage values are with respect to network ground terminal.
5. C_L includes probe and jig capacitance.
6. All diodes are 1N3064, or equivalent.
7. Inputs and outputs not specified are open.

FIGURE 7. Switching time test circuit - device type 02 - Continued.

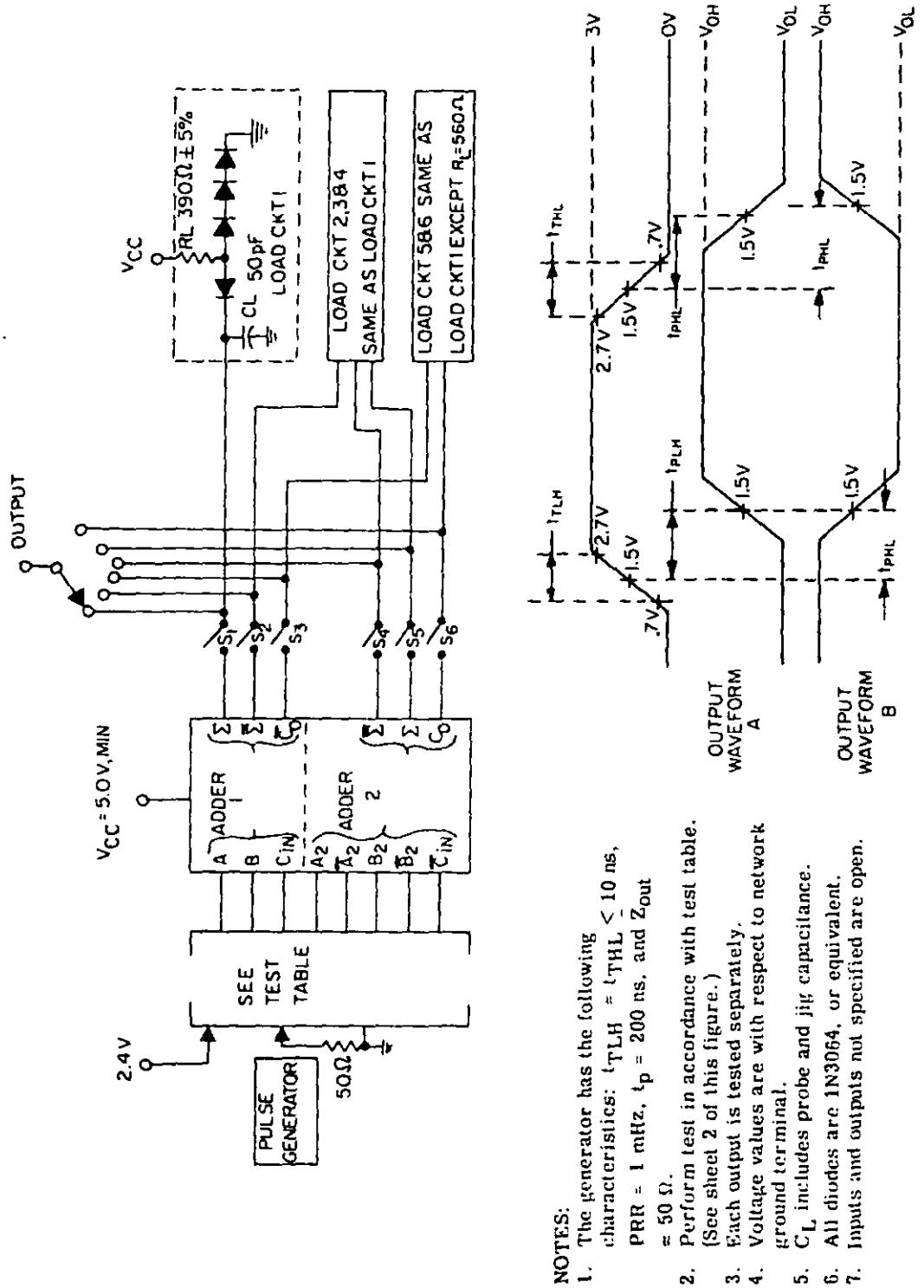
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Switching time set up

Parameter	Apply pulse generator output to	Output under test	Apply 2.4 V to	Apply GND to	S1	S2	S3	S4	S5	S6
t_{PLH1} t_{PHL1}	C_1	(Adder 1) $\Sigma 1$ (Wave form) B	A_1	B_1	Closed	Closed	Closed	Open	Open	Open
t_{PLH2} t_{PHL2}	C_1	(Adder 1) $\Sigma 1$ (Wave form) A	A_1	B_1	Closed	Closed	Closed	Open	Open	Open
t_{PLH3} t_{PHL3}	C_1	(Adder 1) C_{O1} (Wave form) B	A_1	B_1	Closed	Closed	Closed	Open	Open	Open
t_{PLH4} t_{PHL4}	A_2	(Adder 2) $\Sigma 2$ (Wave form) A	C_2 $\bar{A}_2$, $\bar{C}_{IN2}$	B_2, $\bar{B}_2$	Open	Open	Open	Closed	Closed	Closed
t_{PLH5} t_{PHL5}	A_2	(Adder 2) $\Sigma 2$ (Wave form) B	C_2 $\bar{A}_2$, $\bar{C}_{IN2}$	B_2, $\bar{B}_2$	Open	Open	Open	Closed	Closed	Closed
t_{PLH6} t_{PHL6}	A_2	(Adder 2) C_{O2} (Wave form) A	C_2 $\bar{A}_2$, $\bar{C}_{IN2}$	B_2, $\bar{B}_2$	Open	Open	Open	Closed	Closed	Closed

*FIGURE 8. Switching time test circuit - device type 03.

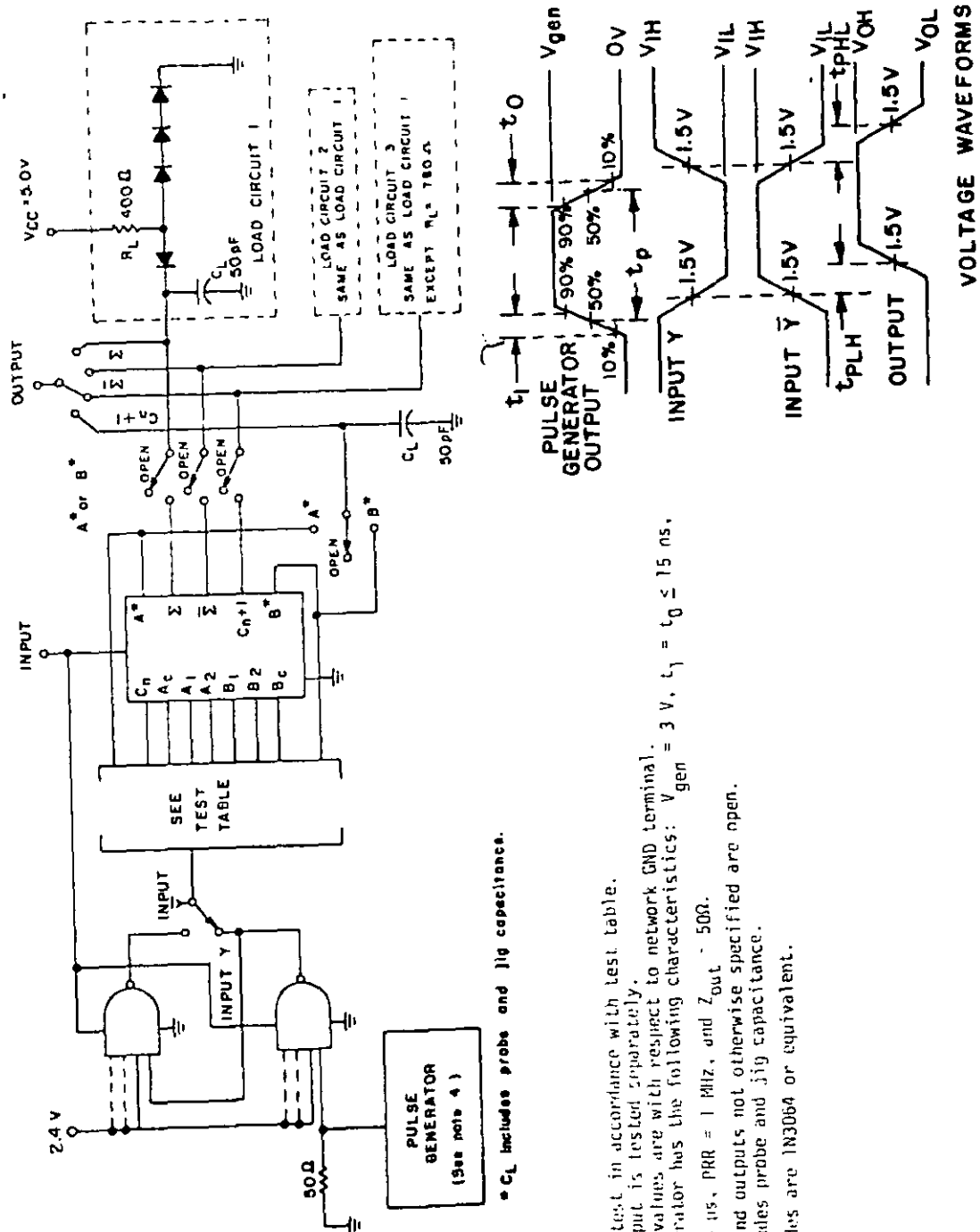
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NOTES:
1. The generator has the following characteristics: $t_{TLH} = t_{THL} < 10$ ns, PRR = 1 mHz, t_p = 200 ns, and Z_{out} = 50 Ω.
2. Perform test in accordance with test table. (See sheet 2 of this figure.)
3. Each output is tested separately.
4. Voltage values are with respect to network ground terminal.
5. C_L includes probe and jig capacitance.
6. All diodes are 1N3064, or equivalent.
7. Inputs and outputs not specified are open.

*FIGURE 8. Switching time test circuit - device type 03 - Continued.

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NOTES:
1. Perform test in accordance with test table.
2. Each output is tested separately.
3. Voltage values are with respect to network GND terminal.
4. The generator has the following characteristics: V_{gen} = 3 V, $t_1 = t_0 \leq 15$ ns, t_p = 0.5 us, PRR = 1 MHz, and Z_{out} = 50Ω.
5. Inputs and outputs not otherwise specified are open.
6. C_L includes probe and jig capacitance.
7. All diodes are 1N3064 or equivalent.

FIGURE 9. Switching time test circuit - device type 04.

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TEST TABLE (SEE NOTE 5)

TEST NO.	SYMBOL	OUTPUT UNDER TEST	APPLY INPUT Y TO	APPLY INPUT Y TO	APPLY +2.4V TO	APPLY GND TO	APPLY OUTPUT LOADING TO
64, 76	t_{PHL1}	$\overline{C_{n+1}}$	NONE	C_n	NONE	B_1	$\overline{C_{n+1}}$ (Load circuit 3)
65, 77	t_{PLH1}	C_{n+1}	NONE	C_n	NONE	B_1	$\overline{C_{n+1}}$ (Load circuit 3)
66, 78	t_{PHL2}	$\overline{C_{n+1}}$	B_c	NONE	C_n	A_1, B_1	$\overline{C_{n+1}}$ (Load circuit 3)
67, 79	t_{PLH2}	$\overline{C_{n+1}}$	B_c	NONE	C_n	A_1, B_1	$\overline{C_{n+1}}$ (Load circuit 3)
68, 80	t_{PHL3}	Σ	A_c	NONE	C_n	A_1, B_1	$\overline{\Sigma}$ (Load circuit 2) Σ (Load circuit 1) $\overline{C_{n+1}}$ (Load circuit 3)
69, 81	t_{PLH3}	Σ	A_c	NONE	C_n	A_1, B_1	Σ (Load circuit 1) $\overline{\Sigma}$ (Load circuit 2) $\overline{C_{n+1}}$ (Load circuit 3)
70, 82	t_{PHL4}	$\overline{\Sigma}$	B_c	NONE	C_n	B_1	$\overline{\Sigma}$ (Load circuit 2)
71, 83	t_{PLH4}	$\overline{\Sigma}$	B_c	NONE	C_n	B_1	$\overline{\Sigma}$ (Load circuit 2)
72, 84	t_{PHL5}	A^*	NONE	A_1	A_2	NONE	A^* C_L = 50 pF
73, 85	t_{PLH5}	A^*	NONE	A_1	A_2	NONE	A^* C_L = 50 pF
74, 86	t_{PHL6}	B^*	NONE	B_1	B_2	NONE	B^* C_L = 50 pF
75, 87	t_{PLH6}	B^*	NONE	B_1	B_2	NONE	B^* C_L = 50 pF

FIGURE 9. Switching time test circuit - device type 04 - Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be H ≥ 2.0 V; L ≤ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14				
			Test No.	Σ1	A1	B1	VCC	C0	NC	NC	NC	NC	C2	GND	Σ2	B2	A2		Min	Max	Unit
1	V_{IC}		1		-12mA		4.5 v							GND				A1	---	-1.5	V
T_A = 25°C			2			-12mA												B1		-1.5	
			3															A2		-1.5	
			4													-12mA	-12mA	B2		-1.5	
			5					-12mA										C0		-1.5	
	V_{OH}	3006	6	-0.4 mA	2.0 v	2.0 v		2.0 v							-0.4 mA	2.0 v	2.0 v	Σ1	2.4		
			7										-0.2 mA					Σ2			
			8															C2			
	V_{OL}	3007	9	16 mA	0.8 v	0.8 v		0.8 v							16 mA	0.8 v	0.8 v	Σ1	---	0.4	
			10					0.8 v					8 mA					Σ2			
			11					0.8 v										C2			
	I_{IH_1}	3010	12		GND	GND	5.5 v	2.4 v										C0		100	µA
			13		2.4 v	GND		GND										A1			
			14		GND	2.4 v		GND										B1			
			15													2.4 v	2.4 v	A2			
			16															B2			
	I_{IH_2}		17		GND	GND		5.5 v										C0		400	
			18		5.5 v	GND		GND										A1			
			19		GND	5.5 v		GND										B1			
			20														5.5 v	A2			
			21													5.5 v		B2			
	I_{IL_1}	3009	22 CKT A		0.4 v	5.5 v		5.5 v										A1	-2.1	-6.4	mA
			22 CKT B		0.4 v	5.5 v		5.5 v										A1	-1.4	-3.2	
			23 CKT A		5.5 v	0.4 v		5.5 v										B1	-2.1	-6.4	
			23 CKT B			0.4 v		5.5 v										B1	-1.4	-3.2	
			24 CKT A			5.5 v		0.4 v										C0	-2.1	-6.4	
			24 CKT B			5.5 v		0.4 v										C0	-1.4	-3.2	
	I_{IL_2}		25 CKT A														0.4 v	A2	-0.7	-1.6	
			25 CKT B														0.4 v	A2	-1.4	-3.2	
			26 CKT A													0.4 v		B2	-0.7	-1.6	
			26 CKT B													0.4 v		B2	-1.4	-3.2	
	I_{OS_1}	3011	27 CKT A	GND	4.5 v	4.5 v		4.5 v								4.5 v	4.5 v	Σ1	-20	-55	
			27 CKT B	GND											GND			Σ1	-20	-70	
			28 CKT A												GND			Σ2	-20	-55	
			28 CKT B															Σ2	-20	-70	
	I_{OS_2}		29										GND					C2	-20	-70	
	I_{CC}	3005	30 CKT A			GND										GND	4.5 v	VCC	---	50	
			30 CKT B			GND										GND	4.5 v	VCC	---	40	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55 C and V_{IC} tests are omitted.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H = 2.0 V; L = 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits		
			Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Unit
			Test No.	Σ1	A1	B1	VCC	C0	NC	NC	NC	NC	C2	(GND)	Σ2	B2	A2				
7 2/ T_A = 25°C			31	L	0.8v 3/	0.8v 3/	4.5 v	0.8v 3/					L		L	0.8v 3/	0.8v 3/	All outputs	H or L as shown 1/		
			32	H	2.0 v	0.8 v									L						
			33	H	0.8 v	2.0 v									L						
			34	L	2.0 v	2.0 v									H						
			35	L	0.8 v	0.8 v									H		2.0 v				
			36	H	2.0 v	0.8 v									H						
			37	H	0.8 v	2.0 v									H						
			38	L	2.0 v	2.0 v							H		L						
			39	L	0.8 v	0.8 v							L		H	2.0 v	0.8 v				
			40	H	2.0 v	0.8 v							L		H						
			41	H	0.8 v	2.0 v							L		H						
			42	L	2.0 v	2.0 v							H		L						
			43	L	0.8 v	0.8 v									L		2.0 v				
			44	H	2.0 v	0.8 v									L						
			45	H	0.8 v	2.0 v									L						
			46	L	2.0 v	2.0 v									H						
			47	H	0.8 v	0.8 v		2.0 v					L		L	0.8 v	0.8 v				
			48	L	2.0 v	0.8 v									H						
			49	L	0.8 v	2.0 v									H						
			50	H	2.0 v	2.0 v									H						
			51	H	0.8 v	0.8 v									H		2.0 v				
			52	L	2.0 v	0.8 v							H		L						
			53	L	0.8 v	2.0 v							H		L						
			54	H	2.0 v	2.0 v							H		L						
			55	H	0.8 v	0.8 v							L		H	2.0 v	0.8 v				
			56	L	2.0 v	0.8 v							H		L						
			57	L	0.8 v	2.0 v									L						
			58	H	2.0 v	2.0 v									L						
			59	H	0.8 v	0.8 v									L		2.0 v				
			60	L	2.0 v	0.8 v									H						
			61	L	0.8 v	2.0 v									H						
			62	H	2.0 v	2.0 v									H						
8 2/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55°C.																				
9 T_A = 25°C	t_{PLH1}	3003	63	See figure 6.														C0 to Σ1	3	37	ns
	t_{PHL1}		64															C0 to Σ1		43	
	t_{PLH2}		65															B2 to Σ2		43	
	t_{PHL2}		66															B2 to Σ2		38	
	t_{PLH3}		67															C0 to Σ2		42	
	t_{PHL3}		68															C0 to Σ2		45	
	t_{PLH4}		69															C0 to C2		21	
	t_{PHL4}		70															C0 to C2		30	
10 T_A = 125°C	t_{PLH1}		71	See figure 6.														C0 to Σ1	3	56	
	t_{PHL1}		72															C0 to Σ1		61	
	t_{PLH2}		73															B2 to Σ2		65	
	t_{PHL2}		74															B2 to Σ2		54	
	t_{PLH3}		75															C0 to Σ2		62	
	t_{PHL3}		76															C0 to Σ2		61	
	t_{PLH4}		77															C0 to C2		34	
	t_{PHL4}		78															C0 to C2		43	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = 55°C.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be H $\geq$ 2.0 V; L $\leq$ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Max	Unit
				A4	Σ3	A3	B3	VCC	Σ2	B2	A2	Σ1	A1	B1	GND	C0	C4	Σ4	B4				
1 T_A = 25°C	V_{OH}	3006	1	2.0 v	-0.4mA	2.0 v	2.0 v	4.5 v	-0.4mA	2.0 v	2.0 v	-0.4mA	2.0 v	2.0 v	GND	2.0 v	-0.2mA	-0.4mA	2.0 v	Σ1 Σ2 Σ3 Σ4 C4	2.4	---	V
	V_{OL}	3007	6	0.8 v	16 mA	0.8 v	0.8 v		16 mA	0.8 v	0.8 v	16mA	0.8 v	0.8 v		0.8 v	8 mA	16 mA	0.8 v	Σ1 Σ2 Σ3 Σ4 C4	---	0.4	
	V_{IC}		11	-12 mA		-12 mA	-12 mA			-12 mA	-12 mA		-12 mA	-12 mA		-12 mA			-12 mA	A4 A3 B3 B2 A2 A1 B1 C0 B4		-1.5	
	I_{IH_1}	3010	20	2.4 v		2.4 v GND	GND 2.4 v	5.5 v		GND GND 2.4 v 5/	GND GND 5/ 2.4 v		2.4 v GND GND6/	GND 2.4 v GND6/		GND GND 2.4 v			5/ 2.4 v	A3 B3 A1 B1 C0 A4 B2 A2 B4		160 40	μA
	I_{IH_2}	3010	29	5.5 v 5/		5.5 v GND	GND 5.5 v			GND GND 5.5 v 5/	GND GND 5/ 5.5 v		5.5 v GND GND	GND 5.5 v GND		GND GND 5.5 v			5/ 5.5 v	A3 B3 A1 B1 C0 A4 B2 A2 B4		400	

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V; L ≤ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Test limits Max	Unit
			Test No.	A4	Σ3	A3	B3	VCC	Σ2	B2	A2	Σ1	A1	B1	GND	C0	C4	Σ4	B4				
1		3009	38 CKT C,G,H			0.4 v	0.8 v	5.5 v		5.5 v	5.5 v				GND					A3	-2.3	-6.4	mA
T_A 25 C	I_{IL1}		38 CKT D			0.4 v	5.5 v													A3	-1.4	-3.2	
			38 CKT E			0.4 v	5.5 v													A3	-0.8	-2.6	
			38 CKT F			0.4 v	5.5 v													A3	-1.4	-3.2	
			39 CKT C,G,H			0.8 v	0.4 v													B3	-2.3	-6.4	
			39 CKT D			5.5 v	0.4 v													B3	-1.4	-3.2	
			39 CKT E			5.5 v	0.4 v													B3	-0.8	-2.6	
			39 CKT F			5.5 v	0.4 v													B3	-1.4	-3.2	
			40 CKT C,G,H										0.4 v	0.8 v		5.5 v				A1	-2.3	-6.4	
			40 CKT D										0.4 v	5.5 v						A1	-1.4	-3.2	
			40 CKT E										0.4 v	5.5 v						A1	-0.8	-2.6	
			40 CKT F										0.4 v	5.5 v						A1	-1.4	-3.2	
			41 CKT C,G,H										0.8 v	0.4 v						B1	-2.3	-6.4	
			41 CKT D										5.5 v	0.4 v						B1	-1.4	-3.2	
			41 CKT E										5.5 v	0.4 v						B1	-0.8	-2.6	
			41 CKT F										5.5 v	0.4 v						B1	-1.4	-3.2	
			42 CKT C										5.5 v	0.8 v		0.4 v				C0	-2.1	-6.4	
			42 CKT D										5.5 v	5.5 v		0.4 v				C0	-1.4	-3.2	
			42 CKT E										5.5 v	5.5 v		0.4 v				C0	-0.7	-1.6	
			42 CKT F										5.5 v	5.5 v		0.4 v				C0	-1.4	-3.2	
			42 CKT G, H										5.5 v	5.5 v		0.4 v				C0	-2.1	-4.8	
	I_{IL2}		43 CKT C,G,H	0.4 v																A4	-0.7	-1.6	
			43 CKT D	0.4 v															5.5 v	A4	-1.4	-3.2	
			43 CKT E	0.4 v																A4	-0.8	-2.6	
			43 CKT F	0.4 v																A4	-0.8	-2.6	
			44 CKT C,G,H							0.4 v										B2	-0.7	-1.6	
			44 CKT D							0.4 v	5.5 v									B2	-1.4	-3.2	
			44 CKT E							0.4 v										B2	-0.8	-2.6	
			44 CKT F							0.4 v										B2	-0.8	-2.6	
			45 CKT C,G,H								0.4 v									A2	-0.7	-1.6	
			45 CKT D							5.5 v	0.4 v									A2	-1.4	-3.2	
			45 CKT E								0.4 v									A2	-0.8	-2.6	
			45 CKT F								0.4 v									A2	-0.8	-2.6	
			46 CKT C,G,H																0.4 v	B4	-0.7	-1.6	
			46 CKT D	5.5 v															0.4 v	B4	-1.4	-3.2	
			46 CKT E																0.4 v	B4	-0.8	-2.6	
			46 CKT F																0.4 v	B4	-0.8	-2.6	
	I_{OS1} 4/	3011	47	4.5 v	GND	4.5 v	4.5 v		GND	4.5 v	4.5 v	GND	4.5 v	4.5 v		4.5 v		GND	4.5 v	Σ3	-20	-55	
			48																	Σ2			
			49																	Σ1			
			50																	Σ4			
	I_{OS2}		51														GND			C4	-20	-70	
	I_{CC}	3005	52 CKT C, E, G, H	4.5 v		4.5 v	GND			GND	4.5 v		4.5 v	GND		4.5 v			GND	V_{CC}	---	100	
			52 CKT D, F	4.5 v		4.5 v	GND			GND	4.5 v		4.5 v	GND		4.5 v			GND	V_{CC}	---	80	

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Terminal conditions (pins not designated may be H ≥ 2.0 V; L ≤ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	A4	Σ3	A3	B3	VCC	Σ2	B2	A2	Σ1	A1	B1	GND	C0	C4	Σ4	B4		Min	Max	Unit
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																						
7 2/ T_A = 25°C			53	0.8 v 3/	L 1/	0.8 v 3/	0.8 v 3/	4.5 v	L 1/	0.8 v 3/	0.8 v 3/	L 1/	0.8 v 3/	0.8 v 3/	GND	0.8 v 3/	L 1/	L 1/	0.8 v 3/	All outputs	H or L as shown 1/		
			54		H	2.0 v	0.8 v		L			H	0.8 v	2.0 v				L					
			55		H	0.8 v	2.0 v		L			H	2.0 v	2.0 v				L					
			56		L	2.0 v	2.0 v		H		2.0 v	L	0.8 v	0.8 v				H					
			57	2.0 v	L	0.8 v	0.8 v		H			H	2.0 v	0.8 v				L					
			58		H	2.0 v	0.8 v		H			H	0.8 v	2.0 v				H					
			59		H	0.8 v	2.0 v		L			L	2.0 v	2.0 v				L	2.0 v				
			60		H	2.0 v	2.0 v		L	2.0 v	0.8 v	L	0.8 v	0.8 v				H					
			61	0.8 v	L	0.8 v	0.8 v		H			H	2.0 v	0.8 v				L					
			62		H	2.0 v	0.8 v		H			H	0.8 v	2.0 v				H					
			63		H	0.8 v	2.0 v		L			L	2.0 v	2.0 v				L					
			64		H	2.0 v	2.0 v		L		2.0 v	L	0.8 v	0.8 v				L					
			65	2.0 v	H	0.8 v	0.8 v		L			H	2.0 v	0.8 v				H					
			66		L	2.0 v	0.8 v		L			H	0.8 v	2.0 v				L					
			67		L	0.8 v	2.0 v		L			L	2.0 v	2.0 v				L					
			68		H	2.0 v	2.0 v		L			H	0.8 v	0.8 v		2.0 v		L		0.8 v			
			69	0.8 v	L	0.8 v	0.8 v		L	0.8 v	0.8 v	L	2.0 v	0.8 v				L					
			70		H	2.0 v	0.8 v		L			L	0.8 v	2.0 v				L					
			71		H	0.8 v	2.0 v		L			H	2.0 v	2.0 v				L					
			72		L	2.0 v	2.0 v		L		2.0 v	H	0.8 v	0.8 v				H		2.0 v			
			73	2.0 v	L	0.8 v	0.8 v		L			L	2.0 v	0.8 v				H					
			74		L	2.0 v	0.8 v		L			L	0.8 v	2.0 v				H					
			75		L	0.8 v	2.0 v		L			H	2.0 v	2.0 v				H					
			76		H	2.0 v	2.0 v		L			H	0.8 v	0.8 v				H					
			77	0.8 v	L	0.8 v	0.8 v		L	2.0 v	0.8 v	L	2.0 v	0.8 v				H					
			78		L	2.0 v	0.8 v		L			L	0.8 v	2.0 v				H					
			79		L	0.8 v	2.0 v		L			H	2.0 v	2.0 v				H					
			80		H	2.0 v	2.0 v		L		2.0 v	H	0.8 v	0.8 v				H					
			81	2.0 v	H	0.8 v	0.8 v		L			L	2.0 v	0.8 v				H					
			82		L	2.0 v	0.8 v		H			L	0.8 v	2.0 v				H					
			83		L	0.8 v	2.0 v		H			H	2.0 v	2.0 v				H					
			84		H	2.0 v	2.0 v		H			H											
8 2/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55°C.																						

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Terminal conditions (pins not designated may be H ≥ 2.0 V; L ≤ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	A4	Σ3	A3	B3	VCC	Σ2	B2	A2	Σ1	A1	B1	GND	C0	C4	Σ4	B4		Min	Max	Unit
9	t_{PLH1}	3003	85																	C0 to Σ1	3	37	ns
T_A = 25°C	t_{PHL1}		86																	C0 to Σ1		44	
	t_{PLH2}		87																	B2 to Σ2		41	
	t_{PHL2}		88																	B2 to Σ2		36	
	t_{PLH3}		89																	C0 to Σ3		42	
	t_{PHL3}		90																	C0 to Σ2		47	
	t_{PLH4}		91																	C0 to Σ3		55	
	t_{PHL4}		92																	C0 to Σ3		66	
	t_{PLH5}		93																	C0 to Σ4		61	
	t_{PHL5}		94		See figure 7.															C0 to Σ4		61	
	t_{PLH6}		95																	C0 to C4		54	
	t_{PHL6}		96																	C0 to C4		40	
	t_{PLH7}		97																	A2 to Σ2		41	
	t_{PHL7}		98																	A2 to Σ2		36	
	t_{PLH8}		99																	A4 to Σ4		41	
	t_{PHL8}		100																	A4 to Σ4		36	
	t_{PLH9}		101																	B4 to Σ4		41	
	t_{PHL9}		102																	B4 to Σ4		36	
10	t_{PLH1}		103																	C0 to Σ1	3	39	
T_A = 125°C	t_{PHL1}		104																	C0 to Σ1		51	
	t_{PLH2}		105																	B2 to Σ2		45	
	t_{PHL2}		106																	B2 to Σ2		40	
	t_{PLH3}		107																	C0 to Σ2		51	
	t_{PHL3}		108																	C0 to Σ2		54	
	t_{PLH4}		109																	C0 to Σ3		62	
	t_{PHL4}		110																	C0 to Σ3		77	
	t_{PLH5}		111		See figure 7.															C0 to Σ4		72	
	t_{PHL5}		112																	C0 to Σ4		71	
	t_{PLH6}		113																	C0 to C4		63	
	t_{PHL6}		114																	C0 to C4		55	
	t_{PLH7}		115																	A2 to Σ2		45	
	t_{PHL7}		116																	A2 to Σ2		40	
	t_{PLH8}		117																	A4 to Σ4		45	
	t_{PHL8}		118																	A4 to Σ4		40	
	t_{PLH9}		119																	B4 to Σ4		45	
	t_{PHL9}		120																	B4 to Σ4		40	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55°C.																						

1/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H ≥ 1.5 V and L ≤ 1.5 V when using a high speed checker single comparator.
2/ Only a summary of attributes data is required.
3/ Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
4/ Circuit B limits for I_{OS} shall be - 70 mA, max.
5/ "GND" when testing for circuit D, E, F, G and H.
6/ Substitute "Open" for circuit F.

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TABLE III. Group A inspection for device type 03.

Terminal conditions (pins not designated may be H ≥ 2.0 V; L ≤ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 $\overline{A2}$	2 A1	3 B1	4 C1	5 $\overline{C01}$	6 $\overline{\Sigma1}$	7 $\Sigma1$	8 GND	9 $\overline{\Sigma2}$	10 $\Sigma2$	11 C02	12 $\overline{C2}$	13 B2	14 A2	15 $\overline{B2}$	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{OH}	3006	1		2.0 v	0.8 v	0.8 v	-560µA	-800 µA		GND								4.5 v	$\overline{C01}$	2.4		V
			2		0.8 v	0.8 v	0.8 v													$\overline{\Sigma1}$			
			3		2.0 v	2.0 v	2.0 v			-800 µA										$\Sigma1$			
			4	0.8 v									-800 µA	-560 µA	2.0 v	0.8 v	0.8 v	0.8 v		C02			
			5	2.0 v											0.8 v	2.0 v	2.0 v	2.0 v		$\overline{\Sigma2}$			
			6	2.0 v								-8.00µA			2.0 v	0.8 v	0.8 v	2.0 v		$\overline{\Sigma2}$			
	V_{OL}	3007	7		0.8 v	2.0 v	2.0 v	8 mA	16 mA											$\overline{C01}$	---	0.4	
			8		2.0 v	0.8 v	0.8 v													$\overline{\Sigma1}$			
			9		0.8 v	0.8 v	0.8 v			16 mA										$\Sigma1$			
			10	2.0 v									16 mA	8 mA	2.0 v	0.8 v	0.8 v	0.8 v		C02			
			11	2.0 v											0.8 v	2.0 v	2.0 v	2.0 v		$\Sigma2$			
			12	0.8 v								16 mA			0.8 v	0.8 v	0.8 v	0.8 v		$\overline{\Sigma2}$			
	I_{IL1}	3009	13													0.4 v			5.5 v	B2	-0.4	-1.3	mA
	I_{IL1}		14														0.4 v			A2	-0.4	-1.3	
	I_{IL2}		15 CKT J		0.4 v	4.5 v	4.5 v													A1	-1.2	-5.6	
			15 CKT K		0.4 v	4.5 v	4.5 v													A1		-3.9	
			16 CKT J		4.5 v	0.4 v	4.5 v													B1		-5.6	
			16 CKT K		4.5 v	0.4 v	4.5 v													B1		-3.9	
			17 CKT J		4.5 v	4.5 v	0.4 v													C1		-5.6	
			17 CKT K		4.5 v	4.5 v	0.4 v													C1		-3.9	
	I_{IL3}		18 CKT J	0.4 v											4.5 v	GND	GND	4.5 v		$\overline{A2}$	-2.4	-5.6	
			18 CKT K	0.4 v											4.5 v			4.5 v		$\overline{A2}$	-1.2		
			19 CKT J	4.5 v											0.4 v			4.5 v		$\overline{C2}$	-2.4		
			19 CKT K	4.5 v											0.4 v			4.5 v		$\overline{C2}$	-1.2		
			20 CKT J	4.5 v											4.5 v			0.4 v		$\overline{B2}$	-2.4		
			20 CKT K	4.5 v											4.5 v			0.4 v		$\overline{B2}$	-1.2		
	I_{IH1}	3010	21	GND	GND	GND	GND								GND	2.4 v	GND			B2	---	40	µA
	I_{IH1}		22	GND	GND	GND	GND								GND	GND	2.4 v			A2		40	µA
	I_{IH3}		23	2.4 v	GND	GND	GND								GND	GND	GND	GND		$\overline{A2}$		160	µA
			24	GND	2.4 v	GND														A1			
			25		GND	2.4 v														B1			
			26			GND	2.4 v													C1			
			27				GND													$\overline{C2}$			
			28				GND								2.4 v			2.4 v		$\overline{B2}$			
	I_{IH4}		29	5.5 v	GND	GND	GND								GND	GND	GND	GND		$\overline{A2}$		400	µA
			30	GND	5.5 v	GND														A1			
			31		GND	5.5 v														B1			
			32			GND	5.5 v													C1			
			33				GND													$\overline{C2}$			
			34				GND								5.5 v			5.5 v		$\overline{B2}$			
	I_{OS}	3011	35	GND	GND	GND	GND	GND	GND											$\overline{C01}$	-30	-100	mA
			36		GND															$\overline{\Sigma1}$			
			37		4.5 v					GND										$\Sigma1$			
			38		GND								GND	GND	GND	GND	GND	GND		$\overline{C2}$			
			39															GND		$\Sigma2$			
			40									GND						4.5 v		$\overline{\Sigma2}$			

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be H = 2.0 V, L = 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 $\overline{A2}$	2 A1	3 B1	4 C1	5 $\overline{C01}$	6 $\overline{S1}$	7 S1	8 GND	9 $\overline{S2}$	10 S2	11 C02	12 $\overline{C2}$	13 B2	14 A2	15 $\overline{B2}$	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25 C	V_{IC}		41	12 mA							GND								4.5 v	$\overline{A2}$		1.5	V
			42		-12 mA															A1			
			43			-12 mA														B1			
			44				-12 mA													C1			
			45												-12 mA					$\overline{C2}$			
			46													-12 mA				B2			
			47																	A2			
			48														-12 mA			$\overline{B2}$			
	ICC	3005	49													GND	GND	5.5 v		V_{CC}		55	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125 C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55 C and V_{IC} tests are omitted.																						
7 1/2/ T_A = 25 C			50	0.8 v	0.8 v	0.8 v	0.8 v	H	H	L	GND	L	H	H	0.8 v	0.8 v	0.8 v	0.8 v	4.5 v	All outputs	H or L as shown		
			51	2.0 v	2.0 v	0.8 v	0.8 v	H	L	H		H	L	H				0.8 v					
			52	0.8 v	0.8 v	2.0 v	0.8 v	H	L	H		H	L	H				2.0 v					
			53	2.0 v	2.0 v	2.0 v	0.8 v	L	H	L		L	H	L				2.0 v					
			54	0.8 v	0.8 v	0.8 v	2.0 v	H	L	H		L	H	H			2.0 v	0.8 v					
			55	2.0 v	2.0 v	0.8 v	2.0 v	L	H	L		L	H					0.8 v					
			56	0.8 v	0.8 v	2.0 v	2.0 v	L	H	L		H	L					2.0 v					
			57	*2.0 v*	*2.0 v*	*2.0 v*	*2.0 v*	*L*	*L*	*H*		*H*	*L*					*2.0 v*					
			58	0.8 v								L	H			2.0 v	0.8 v	0.8 v					
			59	2.0 v								H	L					0.8 v					
			60	0.8 v								L	H					2.0 v					
			61	2.0 v								H	L					2.0 v					
			62	0.8 v								L	H				2.0 v	0.8 v					
			63	2.0 v								L	*H*					*0.8 v*					
			64	0.8 v								L	H					2.0 v					
			65	2.0 v								L	H					2.0 v					
			66	0.8 v								H	L		2.0 v	0.8 v	0.8 v	0.8 v					
			67	2.0 v								L	H	L				0.8 v					
			68	0.8 v								L	H	L				*2.0 v*					
			69	2.0 v								H	L	L				2.0 v					
			70	0.8 v								H	L	H			2.0 v	0.8 v					
			71	2.0 v								H	L	H				0.8 v					
			72	0.8 v								L	H	L				2.0 v					
			73	2.0 v								L	H	L				2.0 v					
			74	0.8 v								H	L	H		2.0 v	0.8 v	0.8 v					
			75	2.0 v								L	H	L				0.8 v					
			76	0.8 v								H	L	H				2.0 v					
			77	2.0 v								L	H	L			2.0 v	2.0 v					
			78	0.8 v								H	L	*H*				*0.8 v*					
			79	2.0 v								H	L	*H*				0.8 v					
			80	0.8 v								H	L	H				2.0 v					
			81	2.0 v								H	L	H				2.0 v					
8 1/2/	Same tests, terminal conditions and limits as for subgroup 7, except T_A = 125 and -55°C.																						

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V; L ≤ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 $\overline{A2}$	2 A1	3 B1	4 C1	5 $\overline{C01}$	6 $\overline{\Sigma1}$	7 Σ1	8 GND	9 $\overline{\Sigma2}$	10 Σ2	11 C02	12 $\overline{C2}$	13 B2	14 A2	15 $\overline{B2}$	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_A 25°C	t_{PLH1}	3003	82	Adder 1						See figure 8.										C1 to $\Sigma1$	9	45	ns
	t_{PHL1}		83																	C1 to $\overline{\Sigma1}$	9	48	
	t_{PLH2}		84																	C1 to $\overline{\Sigma1}$	6	31	
	t_{PHL2}		85																	C1 to $\overline{\Sigma1}$	6	31	
	t_{PLH3}		86																	C1 to $\overline{C01}$	3	14	
	t_{PHL3}		87																	C1 to $\overline{C01}$	3	17	
	t_{PLH4}		88	Adder 2																A2 to $\overline{\Sigma2}$	11	52	
	t_{PHL4}		89																	A2 to $\overline{\Sigma2}$	11	55	
	t_{PLH5}		90																	A2 to $\Sigma2$	8	38	
	t_{PHL5}		91																	A2 to $\Sigma2$	8	38	
	t_{PLH6}		92																	A2 to $\overline{C02}$	5	21	
	t_{PHL6}		93																	A2 to $\overline{C02}$	5	24	
10 T_A 125°C	t_{PLH1}		94	Adder 1						See figure 8.										C1 to $\Sigma1$	9	60	
	t_{PHL1}		95																	C1 to $\overline{\Sigma1}$	9	59	
	t_{PLH2}		96																	C1 to $\overline{\Sigma1}$	6	39	
	t_{PHL2}		97																	C1 to $\overline{\Sigma1}$	6	39	
	t_{PLH3}		98																	C1 to $\overline{C01}$	3	21	
	t_{PHL3}		99																	C1 to $\overline{C01}$	3	21	
	t_{PLH4}		100	Adder 2																A2 to $\overline{\Sigma2}$	11	69	
	t_{PHL4}		101																	A2 to $\overline{\Sigma2}$	11	67	
	t_{PLH5}		102																	A2 to $\Sigma2$	8	49	
	t_{PHL5}		103																	A2 to $\Sigma2$	8	48	
	t_{PLH6}		104																	A2 to $\overline{C02}$	5	29	
	t_{PHL6}		105																	A2 to $\overline{C02}$	5	28	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55°C.																						

1/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H ≥ 1.5 V and L < 1.5 V when using a high speed checker single comparator.
2/ Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be H ≥ 2.0 V; L ≤ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D / Case C Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Min	Test limits Max	Test limits Unit
				11	12	13	14	1	2	3	4	5	6	7	8	9	10				
				AC	B1	B2	V_{CC}	B*	BC	Cn	Cn+1	X	X	GND	A1	A2	A*				
1 T_A = 25°C	V_{OH}	3006	1	0.8 V	0.8 V	2.0 V	4.5 V		2.0 V	0.8 V		-.4mA		GND	0.8 V	0.8 V		Y	2.4	---	Vdc
			2	2.0 V	2.0 V	0.8 V			2.0 V	0.8 V			-.4 mA		2.0 V	0.8 V					
			3	2.0 V	2.0 V	0.8 V			2.0 V	0.8 V	-.2 mA				2.0 V	0.8 V		Cn+1			
			4												0.8 V	0.8 V	-.12mA	A*			
			5		0.8 V	0.8 V		-.12 mA										B*			
	V_{OL}	3007	6	2.0 V	0.8 V	2.0 V			2.0 V	0.8 V		16 mA			0.8 V	2.0 V		Y	---	0.4	
			7	0.8 V	0.8 V	2.0 V			2.0 V	0.8 V			16 mA		0.8 V	0.8 V					
			8	2.0 V	0.8 V	0.8 V			0.8 V	0.8 V	8 mA				2.0 V	2.0 V		Cn+1			
			9												2.0 V	2.0 V	4.8mA	A*			
			10		2.0 V	2.0 V		4.8 mA										B*			
	I_{IH1}	3010	11				5.5 V								2.4 V	GND		A1		15	µA
			12												GND	2.4 V		A2			
			13		2.4 V	GND												B1			
			14		GND	2.4 V												B2			
			15	2.4 V													GND	AC			
			16					GND	2.4 V									BC			
	I_{IH3}		17		GND	GND				2.4 V					GND	GND		Cn		200	
	I_{IH4}		18												5.5 V	GND		A1		1	mA
			19												GND	5.5 V		A2			
			20		5.5 V	GND												B1			
			21		GND	5.5 V												B2			
			22	5.5 V													GND	AC			
			23					GND	5.5 V									BC			
			24		GND	GND				5.5 V					GND	GND		Cn			
	I_{IL1}	3009	25												0.4 V	5.5 V		A1		-1.8	mA
			26												5.5 V	0.4 V		A2			
			27		0.4 V	5.5 V												B1			
			28		5.5 V	0.4 V												B2			
			29	0.4 V											GND	GND		AC			
			30		GND	GND			0.4 V									BC			
	I_{IL2}		31	4.5 V											GND	GND	0.4 V	A*		-2.6	
	I_{IL2}		32		GND	GND		0.4 V	4.5 V									B*		-2.6	
	I_{IL3}		33							0.4 V								Cn		-8.0	
	I_{OS1}	3011	34		GND	GND				4.5 V		GND			GND	GND		Y	-20	-57	mA
	I_{OS1}		35							GND			GND						-20	-57	
	I_{OS2}		36								GND							Cn+1	-20	-70	
	I_{OS3}		37														GND	A*	-0.9	-2.9	
	I_{OS3}		38		GND	GND		GND										B*	-0.9	-2.9	
	I_{CC}	3005	39															V_{CC}	---	31	
	V_{IC}		40	-12 mA			4.5 V											AC		-1.5	Vdc
			41												-12 mA			A1			
			42													-12 mA		A2			
			43						-12 mA									BC			
			44		-12 mA													B1			
			45			-12 mA												B2			
			46							-12 mA								Cn			
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125 C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55 C and V_{IC} tests are omitted.																				

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V; L ≤ 0.8 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	11	12	13	14	1	2	3	4	5	6	7	8	9	10		Min	Max	Unit
			Test No.	AC	B1	B2	V_{CC}	B*	BC	Cn	$\overline{Cn+1}$	Σ	$\overline{\Sigma}$	GND	A1	A2	A*				
7 T_A = 25°C	Truth table tests	3014	47	A 1/	B 1/	A	4.5 V		A	B	H	L	H	GND	B	A			See footnote 2		
			48	A	A	B			A	B	H	L	H		A	B					
			49	B	B	A			A	B	H	H	L		B	B					
			50	A	A	B			A	B	H	H	L		A	A					
			51	B	B	B			B	B	H	H	L		B	A					
			52	B	A	A			A	B	H	H	L		A	B					
			53	A	B	B			B	B	L	L	H		A	A					
			54	B	A	A			A	B	L	L	H		B	B					
			55	A	B	A			A	A	H	H	L		B	A					
			56	A	A	B			A	A	H	H	L		A	B					
			57	B	B	A			A	A	L	L	H		B	B					
			58	A	A	B			A	A	L	L	H		A	A					
			59	B	B	B			B	A	L	L	H		B	A					
			60	B	A	A			A	A	L	L	H		A	B					
			61	A	B	B			B	A	L	H	L		A	A					
			62	B	A	A			A	A	L	H	L		B	B					
			63	A	B	B		B	A	A	L	H	L		B	B	B				
8	Same tests, terminal conditions and limits as for subgroup 7, except T_A = -55°C and 125°C.																				
9 T_A = 25°C	t_{PHL1}	3003 (Fig 9)	64		See figure 9													Cn to $\overline{Cn+1}$	---	12	ns
	t_{PLH1}		65															Cn to $\overline{Cn+1}$		22	
	t_{PHL2}		66															BC to Cn+1		60	
	t_{PLH2}		67															BC to Cn+1		30	
	t_{PHL3}		68															AC to Σ		85	
	t_{PLH3}		69															AC to Σ		75	
	t_{PHL4}		70															BC to $\overline{\Sigma}$		80	
	t_{PLH4}		71															BC to $\overline{\Sigma}$		60	
	t_{PHL5}		72															A1 to A*		30	
	t_{PLH5}		73															A1 to A*		70	
	t_{PHL6}		74															B1 to B*		30	
	t_{PLH6}		75															B1 to B*		70	
10 T_A = 125°C	t_{PHL1}	3003 (Fig 9)	76		See figure 9													Cn to $\overline{Cn+1}$		18	ns
	t_{PLH1}		77															Cn to $\overline{Cn+1}$		33	
	t_{PHL2}		78															BC to $\overline{Cn+1}$		90	
	t_{PLH2}		79															BC to Cn+1		45	
	t_{PHL3}		80															AC to Σ		127	
	t_{PLH3}		81															AC to Σ		112	
	t_{PHL4}		82															BC to $\overline{\Sigma}$		120	
	t_{PLH4}		83															BC to $\overline{\Sigma}$		90	
	t_{PHL5}		84															A1 to A*		45	
	t_{PLH5}		85															A1 to A*		105	
	t_{PHL6}		86															B1 to B*		45	
	t_{PLH6}		87															B1 to B*		105	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55°C.																				

1/ A = 2.0 V and B = 0.8 V.

2/ Output voltages shall be either: (a) H = 2.4 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H ≥ 1.5 V and L < 1.5 V when using a high speed checker single comparator.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.3 Ordering data. Procurement documents should specify the following:

(a) Complete part number (see 1.2).
(b) Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
(c) Requirement for certificate of compliance, if applicable.
(d) Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
(e) Requirements for packaging and packing.
(f) Requirements for failure analysis (including required test condition of Method 5003), corrective action and reporting of results, if applicable.
(g) Requirements for product assurance options.
(h) Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, and as follows:

GND - - - - - - - - - - - - - - - Electrical ground (common terminal)
V_{IN} - - - - - - - - - - - - - - - Voltage level at an input terminal
V_{IC} - - - - - - - - - - - - - - - Input clamp voltage
I_{IN} - - - - - - - - - - - - - - - Current flowing into an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. Microcircuits covered by this specification are substitutable for the following commercial device types:

Device type	Commercial type
01	5482, 7482, 9382
02	5483, 7483, 9383
03	9304
04	5480, 7480, 9380

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - EL, MI, MU
Air Force - 11, 19, 99
DSA - ES

User activities:
Army - SM
Navy - CG, MC, AS, OS, SH

Preparing activity:
Air Force - 17

Agent:
DSA - ES

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DEFENSE SUPPLY AGENCY
DoD-304

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

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